



Rich arc routing problem in city logistics: Models and solution algorithms using a fluid queue-based time-dependent travel time representation

Jiawei Lu^a, Qinghui Nie^b, Monirehalsadat Mahmoudi^c, Jishun Ou^{b,*},
Chongnan Li^d, Xuesong Simon Zhou^{a,*}

^a School of Sustainable Engineering and the Built Environment, Arizona State University, Tempe, AZ, 85281, USA

^b College of Architectural Science and Engineering, Yangzhou University, Yangzhou, 225127, China

^c School of Packaging, College of Agriculture and Natural Resources, Michigan State University, East Lansing, Michigan, 48824, USA

^d School of Traffic and Transportation, Beijing Jiaotong University, Beijing, 100044, China

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ABSTRACT

City logistics, as an essential component of the city operation system, aims at managing the complex flow of goods and services from providers to customers efficiently. Delays associated with peak-period traffic congestion exists in both large and small metropolitan areas. As many of the service tasks in city logistics are needed to be performed during peak hours, operators of urban management movement should consider reducing the total trip time and delay when designing service plans. Equally important, the congestion impact of service vehicles to other road users should also be considered. In this paper, we focus on formulating and solving rich arc routing problems (RARPs) in city logistics with a congested urban environment. We highlight the needs of embedding a structurally parsimonious time-dependent travel time model in RARP for producing high-quality and practically useful solutions. A fluid queue model based analytical approach is presented for link travel time calibration in the form of polynomial arrival rate functions. Accordingly, system-wide (societal) impact of vehicles routing is analytically derived and incorporated into the RARP models which enables traffic managers to systematically consider operation costs and societal impacts when designing routing policies in real-life city logistics applications. Additionally, we develop two new representation schemes for time-dependent travel time modeling in RARPs, including a discretized time-expanded representation scheme and a nonlinear polynomial representation scheme. Three modeling approaches for RARPs are proposed, with different perspectives of capturing time-dependent travel time and formulating problem-specific constraints. With a real-life sprinkler truck routing problem as the representative example of RARP, we develop two efficient exact solution algorithms, including a Lagrangian relaxation-based method and a branch-and-price based method. The latter one is embedded with an enhanced parallel branch-and-bound algorithm. Extensive numerical experiments are conducted based on real-world networks and traffic flow data to demonstrate the effectiveness of the proposed methods.

* Corresponding authors.

E-mail addresses: jishun@yzu.edu.cn (J. Ou), xzhou74@asu.edu (X.S. Zhou).

1. Introduction

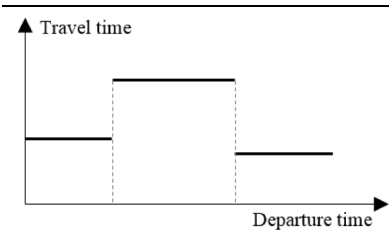
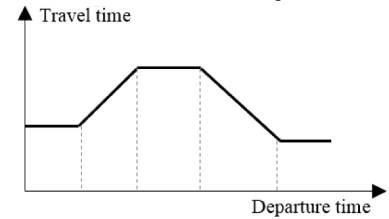
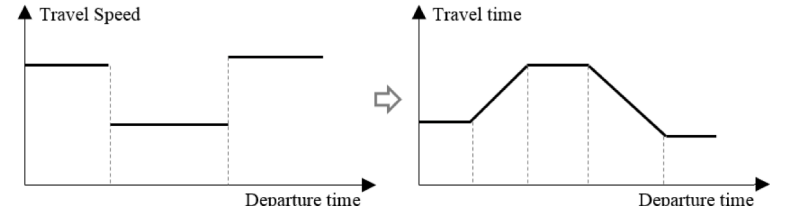
In a broader sense, city logistics refers to the management of the flow of goods and services from providers to customers in urban areas. The urban management movement (UMM) problem (Cattaruzza et al., 2017), as an example, aims to find an optimal set of routes for a fleet of vehicles to satisfy requests for the development, public maintenance, and other functional needs in a city. An efficient city logistics system helps to reduce operation cost, mitigate traffic congestion impact, protect the environment, respond to climate change, connect underserved communities, and support economic vitality. Considering the traffic congestion experienced in cities, this type of problems needs to be formulated as rich arc routing problems (RARPs) under congested traffic conditions.

The word “rich” is used in RARP because besides normal constraints in the variants of the standard ARP (e.g., capacity and time window constraint), some other problem-specific constraints are also considered. For example, in the winter gritting problem, service requests on road links change with time and weather conditions (Eglese and Li, 1992; Eglese, 1994; Li and Eglese, 1996; Tagmouti et al., 2007; Tagmouti et al., 2010; Tagmouti et al., 2011), or in the snow plowing problem, truck routes are specified at the lane level (Perrier et al., 2007a; Perrier et al., 2007b; Salazar-Aguilar et al., 2012; Dussault et al., 2013; Dussault et al., 2014; Quirion-Blais et al., 2017; Castro Campos et al., 2020). Rich constraints considered in RARPs have great practical significance and values while, at the same time, they bring additional challenges, especially for large-scale instances. By fully recognizing rich features in transportation networks, we aim to develop a modeling framework and solution approaches for RARPs that can systematically examine traffic-oriented characteristics.

One of the most important “rich” features of transportation networks is time-varying traffic conditions. Service vehicles may experience time-dependent travel times on roads when serving customers (Liu et al., 2020; Yao et al., 2021). In early research on both vehicle routing problems (VRPs) and ARPs, travel times on links are treated as constant or time-independent; however, in a congested urban environment, solutions obtained with the constant travel time assumption may significantly underestimate the delay and could even lead to infeasibility under tight schedules in real-life applications. Although many recent VRP studies considered piecewise travel time functions to capture time-varying traffic conditions in a more realistic fashion, as discussed by Vidal et al. (2021), additional efforts are still critically needed to offer more precise approximations/representations of reality. On the other hand, in the ARP literature, time-dependent travel times have been largely simplified or ignored (Gendreau et al., 2015). Vidal et al. (2021) first conducted extensive studies on the ARP with time-dependent travel times at a network level and proposed methods for quick travel and service time queries as well as quickest path queries, based on the travel speed function definition given by Ichoua et al. (2003).

Another important goal of RARPs on transportation networks is how to reduce the system-wide (societal) congestion impact of service vehicle routings. RARP applications in city logistics are typically fulfilled by large trucks, e.g., freight trucks and street sweeping trucks. A service truck with a much lower driving speed could affect traffics on multiple lanes which leads to another important class of moving bottleneck problems studied in the literature (see e.g., Li et al., 2020). Thus, RARP applications in city logistics should minimize not only the total operating cost for meeting customer requests but also the potential negative effects to the

Table 1
Three major methods for representing time-dependent travel times in VRP/ARP.

Graphical illustration of methods	Description
	Method type: piecewise-constant travel time function Modeling details: each link has a constant travel time within each pre-defined time period FIFO property: not necessarily satisfied
	Method type: piecewise-linear travel time function Modeling details: each link has a constant travel time within each pre-defined time period; a linear transition line is built between two adjacent periods FIFO property: satisfied
	Method type: piecewise-linear travel time function Modeling details: a piecewise-constant travel speed function is constructed for each link first; piecewise-linear travel time function is then derived based on its corresponding speed function FIFO property: satisfied

background transportation system.

1.1. Related studies

In this section, we review related existing studies along two lines: (1) time-dependent travel time modeling in vehicle routing problems and arc routing problems and (2) rich arc routing problems

1.1.1. Time-dependent travel time modeling in vehicle routing problems and arc routing problems

The important study by Malandraki (1989) first formulated link travel time as a piecewise-constant function of the departure time. Later on, this approach was adopted by Malandraki and Daskin (1992) and Chen et al. (2006). Yet, this approach might not satisfy the First-In-First-Out (FIFO) property. The FIFO property states that, among two identical vehicles that travel along the same path, the one that departs earlier from the origin node could always arrive at the destination node earlier than the other vehicle. Two major approaches are proposed to address the potential FIFO violation issue in the piecewise constant representation. First, piecewise-linear functions are considered by Ahn and Shin (1991) and improved by Fleischmann et al. (2004). In this approach, a linear transition function is built to connect travel times in two adjacent time periods such that travel time changes slowly and smoothly. It can be proved that the FIFO property holds if the slope of linear lines is less than 45° . Another method proposed by Ichoua et al. (2003) starts with time-dependent link travel speed in the form of a piecewise-constant function and derives/computes piecewise-linear travel times satisfying the FIFO property. Table 1 summarizes the above three major methods for time-dependent travel time modeling in VRP and ARP models. Xiao and Konak (2016) offers a more detailed classification and illustration.

Figliozzi (2012) offered a comprehensive set of benchmarks for modeling time-dependent travel times in VRP. The important modeling aspect of time discretization is further studied by Boland et al. (2017) for a broader class of continuous-time shortest path and service network design problems. One can find the related studies in Scherr et al. (2020), Belieres et al. (2021), He et al. (2021), Marshall et al. (2021), Vu et al. (2022), Hewitt (2022), and Lagos et al. (2022). In particular, for the fundamentally important time-dependent shortest path problem, the study by He et al. (2021) systematically considers degree 4 and degree 6 polynomial functions to approximate time-dependent travel time through the piecewise linear interpolants sampled at integer points. The polynomial function can be calibrated for an extended time period (e.g., 24 hours of a day) using real world travel time data, while the related non-linear functional form could have many local minima and maxima, which could greatly affect the computational efficiency of dynamic discretization discovery algorithms (see Boland et al., 2017).

In fact, how to obtain reliable and accurate travel time estimates from field data has remained a challenging problem for the VRP field, while many published papers still use randomly generated or hypothetical travel time distributions for simplicity. By utilizing data from advanced traffic information systems in Berlin, Fleischmann et al. (2004) proposed a smoothing method to ensure the resulting travel time functions satisfy the FIFO property. By using traffic flow data collected from a Belgian highway, Jabali et al. (2009) created a speed profile with five periods, including two periods for morning and afternoon peak hours and three periods for the remaining non-peak hours, for each link within the research area. Kritzing et al. (2012) adopted 15-minute link travel time information from floating car data and developed an extended version of Dijkstra's algorithm to compute the distance matrices between points with various departure times. In the paper by Gmira et al. (2021), the authors developed a discrete-event simulator that generates travel speed updates for real-time vehicle routing applications. Different from the aforementioned numerically driven approximation approaches, Van Woensel et al. (2007) first proposed an innovative queue-theoretic modeling scheme for calibrating expected travel times which was adopted in the studies by Van Woensel et al. (2008) and Lecluyse et al. (2009).

1.1.2. Rich arc routing problems

Considerable efforts have been devoted to the ARP and its variants. Interested readers are referred to a number of survey papers (Wöhlk, 2008; Corberán and Prins, 2010; Corberán and Laporte, 2015; Mourão and Pinto, 2017; Corberán et al., 2021). In this subsection, with the focus on mathematical modeling and solution method development, we review the existing studies along two research lines, including real-life applications of RARP on urban networks and ARP with time-dependent travel times.

(1) Three representative applications of RARP as the urban management movement problem

Sprinkler truck routing problem: Li et al. (2008) investigated the water truck routing problem in open pit mines in which the travel time along a road for each truck is not fixed but relies on its leading truck on the same road. The authors proposed minimum cost flow and set-partitioning based heuristics solution algorithms. Huang and Lin (2014) formulated the street tree watering problem as the periodic arc routing problem with refill points where the watering frequency of each street tree may not be fixed and needs to be scheduled according to the period of watering activity. A graph transformation strategy was first adopted to convert the original problem to a VRP with an ant colony heuristic algorithm. Riquelme-Rodríguez et al. (2014) introduced periodic capacitated arc routing problem with inventory constraints that models the loss of humidity on each road with an inventory consumption function. The quantity of water delivered was considered to be fixed or variable. Two mathematical optimization models were proposed and solved by the commercial solver CPLEX.

Street sweeping problem: Street sweeping requires a special vehicle equipped with a rotating brush that can move along the roadside and sweep material into a container on the vehicle. Early studies (Bodin and Kursh, 1978; Eglese and Murdock, 1991) modeled the street sweeping problem as the capacitated Chinese postman problem and developed heuristics to find reasonable routes for the road-sweeping vehicles. Blazquez et al. (2012) considered two extra “rich” constraints in sweeper route design: the sweepers must visit

each selected street exactly as many times as its number of street sides; certain types of turns are not allowed to use. The original ARP was transformed into a VRP which was further solved by a nearest neighbor heuristic algorithm.

Waste collection problem: Mourão and Almeida (2000) and Mourão and Amado (2005) examined the waste collection problem on randomly generated networks and proposed lower bounds and a three-phase heuristic that transforms one of the lower bound solutions to a feasible one. Maniezzo (2004) considered additional bin compatibility, forbidden turns and one-way street constraints in the model, which is solved by a local search-based heuristic. Ghiani et al. (2005) considered practical constraints in waste collection, e.g., large vehicles are not allowed to use some narrow streets, or services at some sites have to be scheduled at night to avoid traffic congestion. The problem was solved by a cluster-first route-second based heuristic. Many other constraints include traffic regulations (Bautista et al., 2008), mobile depots (Del Pia and Filippi, 2006), intermediate facilities (Ghiani et al., 2001), and trip length restrictions (Ghiani et al., 2010). Recently, Willemse and Joubert (2016a) integrated major problem features of early studies and studied the mixed capacitated arc routing problem with time window and intermediate facilities (MCARPTWIF). Four constructive heuristics were developed and comprehensively evaluated. Given splitting procedures play a key role in giant tour-based heuristics and meta-heuristics, the authors further proposed optimal and heuristic splitting procedures for the MCARPTWIF (Willemse and Joubert, 2016b). In a follow-up study (Willemse and Joubert, 2019), three acceleration mechanisms for local search meta-heuristics were developed to better cope with large-scale instances.

In general, there are three categories of approaches for solving RARPs, including constructive heuristics, meta-heuristics, and exact approaches (Corberán and Laporte, 2015). Constructive heuristics are designed based on problem features and typically provide a single final solution, while meta-heuristics are more general and can be applied to any problems. Both approaches cannot produce measurable quality of solutions. Exact optimization approaches and related model reformulations have not been studied extensively for RARPs, especially with the consideration of time-varying traffic conditions.

(2) ARP with time-dependent travel times

Several VRP-related studies highlight the significant impact of time-dependent travel times on vehicle routing and scheduling. To name a few, Malandraki and Daskin, 1992; Haghani and Jung, 2005; Chen et al., 2006; Donati et al., 2008; Figliozzi, 2012; Dabia et al., 2013; Spliet et al., 2018; and Sun et al., 2018. Specifically, instead of using simplified customer-based graphs, some researchers directly solve VRPs on graphs that are similar to original road networks to incorporate detailed road-network information in the modeling process (Ben Ticha et al., 2021; Huang et al., 2017; Ben Ticha et al., 2019). Ben Ticha et al. (2018) offered a comprehensive review on VRP studies using road-network information.

On the other hand, research on ARP with time-dependent travel times is still limited in the literature. Vidal et al. (2021) recently offered an extensive study for the time-dependent capacitated arc routing problem (TDCARP). Based on the piecewise-constant speed function proposed by Ichoua et al. (2003), the authors derived a closed-form representation for link arrival time functions and developed a continuous preprocessing approach for point-to-point quickest path query. A branch-cut-and-price exact algorithm and a hybrid genetic search-based metaheuristic were proposed for solving the TDCARP.

1.2. Contributions and overview of the paper

In this paper, we focus on model reformulation and developing exact solution approaches for RARPs in congested city transportation network. We have the following observations on the modeling challenges in the related VRP and ARP literature. First, most studies treat travel speeds on roads as constant, while the time-varying feature of transportation networks is largely simplified. A realistic, parsimonious and mathematically rigorous model with a calibration workflow for time-dependent travel time is important for VRP deployment and applications. Second, the system-wide impact of service vehicles in city logistics to the entire transportation system (with other travelers and road users) has not been systematically studied. Third, in most studies, a single model was developed for real-life RARP applications, which lacks a comprehensive investigation and comparison on the effects of rich constraints. Finally, recognizing of the high complexity of RARPs, most studies developed heuristics for solving real-life problems, while exact approaches are critically needed to offer theoretical benchmarks for quantifying the solution quality and the degree of optimality. This paper aims to offer the following contributions to the growing list of modeling and optimization literature for city logistics.

First, recognizing the potential negative impacts of service vehicles to background traffics when providing services, we systematically consider service vehicle operation cost and system (societal) impact of vehicle routings in city logistics so as to reduce the system-wide congestion impact.

Second, based on the fluid queue model, a novel time-dependent travel time representation with the form of nonlinear function is introduced. Compared to the widely adopted piecewise linear functions, the proposed nonlinear travel time function has advantages on parsimonious form, easy calibration, and differentiability. This also leads to the analytical derivations of time-dependent system (societal) impact of service vehicles to background traffics.

Third, three optimization models are developed from different perspectives for modeling RARPs in city logistics, in which the impacts of problem-specific rich constraints on modeling complexity are comprehensively investigated.

Finally, with a real-life sprinkler truck routing problem (SRP) as the representative example of RARP, we develop two exact solution algorithms, namely a Lagrangian relaxation based method and a branch-and-price based method which are embedded with an enhanced parallel branch-and-bound algorithm.

The remainder of this paper is organized as follows. Section 2 presents the methodology for time-dependent travel time modeling and congestion impact analysis on transportation networks. In Section 3, the SRP, as an example of RARPs on transportation networks,

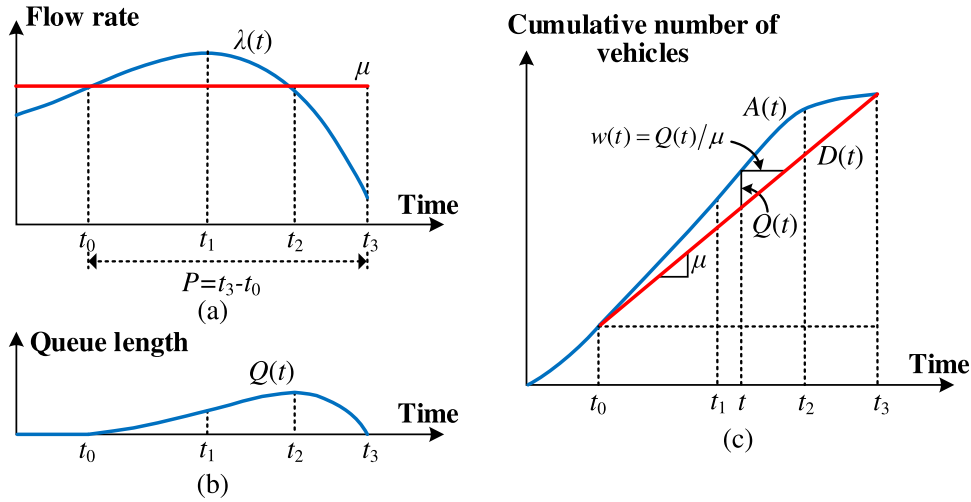


Fig. 1. General graphical illustration of queue evolution for a road link, adapted from Newell (2013), with a single maximum of queue length/-waiting time reached at time t_2 during congestion duration $P = t_3 - t_0$.

is discussed in detail. In Section 4, three mathematical models are proposed for the SRP from different network modeling perspectives, followed by a detailed comparison. Two efficient solution methods, i.e., Lagrangian relaxation and branch-and-price algorithm, are designed in Section 5 for solving the models constructed in Section 4. Comprehensive computational experiments are conducted in Section 6 to evaluate the proposed models and solution approaches.

2. Modeling time-dependent travel time and congestion impacts using fluid queue models with polynomial arrival rates

To calculate the travel time of a specific link, one can divide link length by travel speed, which is in turn derived from simulated/estimated flow/density based on empirically calibrated fundamental diagrams (Greenshields et al., 1935). However, as pointed out by Van Woensel et al. (2007), in the context of vehicle routing problems, instantaneous speed and density have to be embedded in another set of continuum flow models (Kühne and Michalopoulos, 1997) to characterize congestion evolution. As an alternative method, queueing models hold the promise for modeling travel flows with the capability of offering analytical evaluation and sensitivity analysis (Heidemann, 1996). However, the queueing-based approach is mainly based on the stochastic queueing principle with under-saturated conditions such as M/M/1 or G/G/1 (Van Woensel et al., 2007). In this section, by adapting and extending the fluid queue model with quadratic arrival rates proposed by Newell (2013) to represent time-dependent travel times at both link and path levels, we introduce analytical forms that satisfy FIFO conditions and offer precise congestion impact measures during a period of oversaturation. Compared to the widely used piecewise linear functions, the proposed method has the following three advantages: (1) a parsimonious form, (2) an analytical expression with FIFO property, and (3) differentiable.

2.1. Time-dependent travel time

Based on free-flow speed or cut-off speed, which is more precisely defined in congestion bottleneck identification (Hale et al., 2016), traffic states on transportation networks can be classified into two distinct classes: uncongested state and congested state. Under uncongested states or non-peak hours, aggregated vehicle speed on each road link is relatively stable and is approximated using free-flow speed in this paper. This assumption is consistent with the constant free-flow speed in the uncongested regime of triangular fundamental diagram, which is adopted in the widely used cell transmission model (Daganzo, 1994). Under congested states, vehicle travels are constrained by road capacity. Queues form when total inflow travel demand exceeds road capacity. In this case, travel time of vehicles includes two parts: (1) free-flow travel time, and (2) time spent in the queue (travel delay). Due to the dynamics of travel inflow demands, queue length evolves for an extended congestion period, resulting in time-dependent travel time of vehicle trips. In this study, we mainly focus on time-dependent vehicle travel time calibration during each single congested period and use constant or piecewise-linear travel time functions during uncongested periods. It should be noted that, to keep the differentiability of calibrated travel time functions, one can also adopt the proposed single-period calibration method in multi-period applications to avoid connecting adjacent periods with constant functions. To depict potential arrival pattern changes across different periods, high-order arrival rate functions should be applied to reduce modeling errors, but it needs additional calibration efforts (Cheng et al., 2022). Interested readers are also referred to a recent study by Zhou et al. (2022) for a comprehensive investigation on a meso-to-macro cross-resolution performance approach for connecting polynomial arrival queue models to volume-delay functions.

Without loss of generality, each congested road link on transportation networks is modeled as a single queueing system with a constant service rate that equals to the maximum link discharge rate. Fig. 1 shows a graphical illustration of queue evolution for a road link in the classical work by Newell (2013).

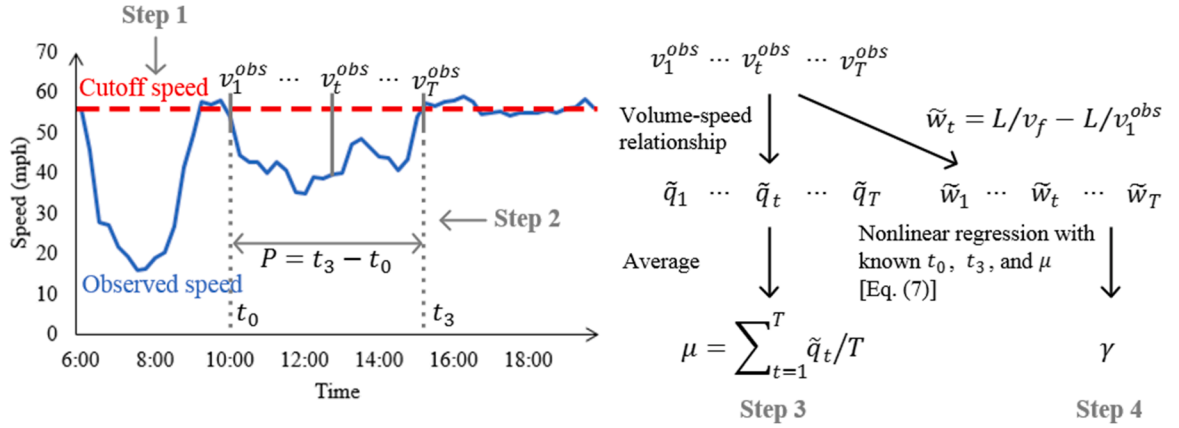


Fig. 2. Graphic illustration of the four-step method for calibrating the key parameters in Eq. (7).

In Fig. 1(a), the blue curve and red straight line represent the arrival rate $\lambda(t)$ and departure (discharge) rate μ , respectively. t_0 , t_2 , and t_3 denote the time at which queue starts to form, queue starts to dissipate, and queue disappears; t_1 is the time with the highest arrival rate. By using the second-order Taylor approximation at time t_1 , arrival rate $\lambda(t)$ can be approximated by the following quadratic function:

$$\lambda(t) = \lambda(t_1) + \lambda'(t_1)(t - t_1) + \frac{1}{2}\lambda''(t_1)(t - t_1)^2. \quad (1)$$

With the observation that $\lambda'(t_1) = 0$, Eq. (1) can be simplified as

$$\lambda(t) = \lambda(t_1) - \gamma(t - t_1)^2, \quad (2)$$

where $\gamma = -\frac{1}{2}\lambda''(t_1)$ ($\gamma > 0$). Notice that $\lambda(t)$ passes two points (t_0, μ) and (t_2, μ) , arrival rate $\lambda(t)$ can also be expressed by the following factored form:

$$\lambda(t) = \gamma(t - t_0)(t_2 - t) + \mu. \quad (3)$$

With Eq. (3), time-dependent queue length $Q(t)$ can be derived as follows:

$$Q(t) = A(t) - D(t) = \int_{t_0}^t [\lambda(\tau) - \mu] d\tau = \gamma(t - t_0)^2 \left[\frac{t_2 - t_0}{2} - \frac{t - t_0}{3} \right], \quad (4)$$

where $A(t)$ and $D(t)$ denote the cumulative arrival and departure at time t , respectively. By introducing the queue clearance time t_3 (i.e., $Q(t_3) = 0$), the following relationship between critical time points can be derived from Eq. (4). The detailed derivation process can be found in Newell (2013) for the quadratic arrival rates and cubic arrival rates by Cheng et al. (2022).

$$t_3 = t_0 + \frac{3}{2}(t_2 - t_0). \quad (5)$$

Then, Eq. (4) can also be written as

$$Q(t) = A(t) - D(t) = \int_{t_0}^t [\lambda(\tau) - \mu] d\tau = \frac{\gamma}{3}(t - t_0)^2(t_3 - t). \quad (6)$$

The discharge rate μ , queue forming time t_0 , and queue clearance t_3 can be directly observed from field data. Thus, time-dependent queue length $Q(t)$ in Eq. (6) only has one inflow demand curvature parameter γ that needs to be calibrated from observed spatial queue length or link travel times. Interested readers can refer to a recent paper by Cheng et al. (2022) for the detailed calibration process that connects the above queuing model with the observations from a spatial queue representation. Finally, by integrating $w(t) = Q(t)/\mu$, time-dependent delay $w(t)$ can be expressed as

$$w(t) = \frac{\gamma}{3\mu}(t - t_0)^2(t_3 - t). \quad (7)$$

In many city logistics applications, only a single data source of observed speed v_t^{obs} is available, where observation time interval t can be 5 minutes or 15 minutes. The following paragraphs describe four steps for calibrating the key parameters of μ and γ in Eq. (7), and the corresponding graphic illustration is provided in Fig. 2.

- (1) Even without the flow count observations, we can still obtain an estimate of the ultimate road capacity c according to the facility types and speed limit. The cutoff speed can be determined based on the well-established traffic fundamental diagram between flow, density and speed, such as Greenshields model (Greenshields et al., 1935).
- (2) One can determine the congestion duration P , while t_0 and t_3 correspond to the timestamps at which speed is dropping from or recovering to the cutoff speed.
- (3) The average discharge rate μ can be estimated from the volume-speed curve and the observed speed during the congestion duration. A default value of μ for undersaturated links can be ultimate road capacity c .
- (4) The space-mean speed v_t^{obs} can be converted to the virtual waiting time $w(t)$ in Eq. (7), and one can use nonlinear regression methods to calibrate parameter γ accordingly. Alternatively, as $w(t_2) = \frac{\gamma}{6\mu}(t_2 - t_0)^3$, we can also quickly estimate γ based on the lowest speed and converted highest waiting time.

It should be noted that, as derivations are based on the point queue model (Vickrey, 1963), $w(t)$ in Eq. (7) represents the delay of a link for vehicles arriving the downstream node (stop line) of the link at time t . With link free-flow travel time $FFTT$, time-dependent travel time of a link during congested periods (denoted by $LT(t)$) can be calculated using Eq. (8) for vehicle leaving from the upstream node of a link at time t .

$$LT(t) = w(t + FFTT) + FFTT = \frac{\gamma}{3\mu}(t + FFTT - t_0)^2(t_3 - t - FFTT) + FFTT. \quad (8)$$

We next derive the range of feasibility for parameter γ in Eq. (8). For a queuing system on transportation networks, the arrival and departure rates must be positive. As we have assumed departure rate μ as a constant parameter, we also require that arrival rate $\lambda(t)$ during the analysis period be nonnegative for any t between t_0 and t_3 . The quadratic function $\lambda(t)$ has the lowest value at time t_3 in the congestion period. That is, $\lambda(t_3) \geq 0$ ensures $\lambda(t) \geq 0$, $\forall t \in [t_0, t_3]$. Integrating Eq. (3) and Eq. (5) yields Eq. (9):

$$\lambda(t_3) = \gamma(t_3 - t_0)(t_2 - t_3) + \mu = \gamma(t_3 - t_0)\left(\frac{2}{3}t_3 + \frac{1}{3}t_0 - t_3\right) + \mu = -\frac{1}{3}\gamma P^2 + \mu \geq 0, \quad (9)$$

where $P = t_3 - t_0$ represents the congestion duration.

Proposition 1. The time-dependent link travel time function $LT(t)$ in Eq. (8) satisfies the FIFO property within the time period of interest $[t_0, t_3]$.

Proof. The FIFO property can be proved if $t + LT(t) \leq t' + LT(t')$ holds for any $t \leq t'$, $t \in [t_0, t_3]$, $t' \in [t_0, t_3]$. Alternatively, it can be proved if $\frac{dLT(t)}{dt} \geq -1$ holds for any $t \in [t_0, t_3]$ (Carey et al., 2014). Based on the derivation of $LT(t)$ given in Eq. (8), $\frac{dLT(t)}{dt}$ can be expressed as

$$\frac{dLT(t)}{dt} = \frac{d\left[\frac{\gamma}{3\mu}(t - t_0)^2(t_3 - t) + FFTT\right]}{dt} = \frac{d\left[\frac{\gamma}{3\mu}(t - t_0)^2(t_3 - t)\right]}{dt}. \quad (10)$$

Let $h = t - t_0$, $h \in [0, P]$, Eq. (10) can be rewritten as

$$\frac{dLT(t)}{dt} = \frac{dLT(h + t_0)}{dh} \frac{dh}{dt} = \frac{d\left[\frac{\gamma}{3\mu}h^2(P - h)\right]}{dh} = \frac{\gamma}{3\mu}(2hP - 3h^2). \quad (11)$$

It is easy to observe that $\frac{\gamma}{3\mu}(2hP - 3h^2)$ is a quadratic function of $h \in [0, P]$ and reaches its minimum at $h = P$. That is,

$$\frac{dLT(t)}{dt} = \frac{\gamma}{3\mu}(2hP - 3h^2) \geq \frac{\gamma}{3\mu}(2PP - 3P^2) = -\frac{\gamma}{3\mu}P^2. \quad (12)$$

By utilizing the range of feasibility for γ derived in Eq. (9), it is obvious that $-\frac{\gamma}{3\mu}P^2 \geq -1$, indicating $\frac{dLT(t)}{dt} \geq -1$, $\forall t \in [t_0, t_3]$. Thus, function $LT(t)$ satisfying the FIFO property is proved.

On the basis of the time-dependent link travel time function, travel time derivations and FIFO property proof are further performed on a path level. Consider two arbitrary nodes i and j on a transportation network, and there are ϕ paths from i to j . Take path $\rho \in \{1, 2, \dots, \phi\}$ as an example, the travel time of path ρ at time t ($PT_\rho(t)$) can be calculated as

$$PT_\rho(t) = \sum_{l=1}^L LT_l(a_l^t), \quad (13)$$

where $l \in \{1, 2, \dots, L\}$ denotes the index of links in path ρ ; L is the total number of links in path ρ ; LT_l represents the link travel time function of the l th link in path ρ ; a_l^t is the arrival time at the upstream node of the l th link if departure from node i at time t . $a_1^t = t$, $a_l^t = a_{l-1}^t + LT_{l-1}(a_{l-1}^t)$ when $2 \leq l \leq L$.

Proposition 2. Path travel time function $PT_\rho(t)$ in Eq. (13) satisfies the FIFO property.

Proof. Consider two vehicles, v^1 and v^2 , departing along the same path ρ at time t and time t' respectively, and $t \leq t'$. That is, $a_1^t \leq a_1^{t'}$. As the travel time function of link 1 satisfies the FIFO property, we have $a_2^t = a_1^t + LT_1(a_1^t) \leq a_2^{t'} = a_1^{t'} + LT_1(a_1^{t'})$. Note that all links

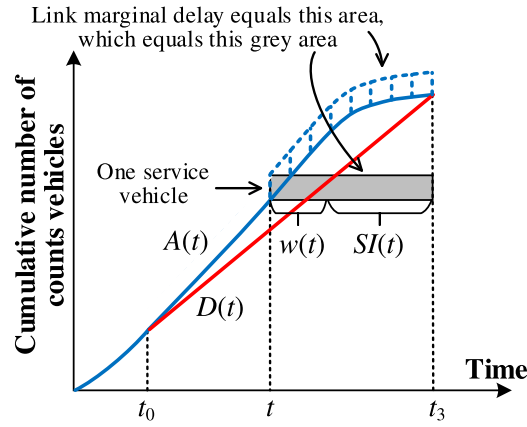


Fig. 3. Illustration of road link marginal delay adapted from Ghali and Smith (1995).

along path ρ satisfy the FIFO property, we can obtain $a_3^t = a_2^t + LT_2(a_2^t) \leq a_3^{t'} = a_2^{t'} + LT_2(a_2^{t'})$, and apply the process recursively till the last link L , i.e., $a_L^t = a_{L-1}^t + LT_{L-1}(a_{L-1}^t) \leq a_L^{t'} = a_{L-1}^{t'} + LT_{L-1}(a_{L-1}^{t'})$. Thanks to the FIFO property on link L , we can easily conclude that vehicle v^1 will arrive at node j earlier than vehicle v^2 . Thus, path travel time $PT_\rho(t)$ satisfying the FIFO property is proved.

It should be noted that the derived path travel time might still have multiple local minima and maxima, but its structurally parsimonious form could balance the tradeoff between computational tractability and the required level of details in representing real-world traffic congestion.

2.2. Analytical form of modeling system-wide congestion impacts to background traffic

As mentioned earlier, service vehicles used in RARP applications are typically slow-moving trucks, thus may bring significant impacts to background traffics, especially during peak hours. In this subsection, we further utilize calibrated queuing profile to analytically measure the system impact of service vehicles, by following the approach proposed by Ghali and Smith (1995).

In Fig. 3, the blue and red solid lines denote the cumulative arrival and departure on a road link, respectively. Let us assume that there is a service truck entering the link at time t . Note that, similar to Section 2.1, we still focus on the congested period, i.e., $t_0 \leq t \leq t_3$. The marginal delay arising from the service truck is the blue dash area, which also equals to the grey area. The marginal delay includes two parts $w(t)$ and $SI(t)$. $w(t)$ is the delay experienced by the service truck, while $SI(t)$ denotes the additional delay experienced on that link by every vehicle arriving between time t and t_3 , due to the arrival of the service truck at time t , which is called system-wide (societal) congestion impact in this paper. Therefore, with the derivation of $w(t)$ in Section 2.1, $SI(t)$ can be calculated as follows:

$$SI(t) = t_3 - t - w(t) = t_3 - t - \frac{\gamma}{3\mu}(t - t_0)^2(t_3 - t). \quad (14)$$

With the consideration that service trucks typically move slower than passenger cars and use more road resources, $SI(t)$ obtained in Eq. (14) further is multiplied by a passenger car equivalent (PCE), i.e., $SI(t) \times PCE$, to measure the system impact of a service truck entering a road link at time t .

With the link-level marginal cost formulation in Eq. (14), system-wide impact along a given path can also be recursively derived similar to travel time derivations in Section 2.1.

3. Sprinkler truck routing: an arc routing application with rich constraints in a congested traffic network

Vehicle travels as well as the action of wind could incur serious airborne particulate matters from roads (Li et al., 2008), causing dust emissions that bring substantial negative effects to surrounding workers and pedestrians. The dust in the air also reduces the visibility of roads and is therefore likely to cause traffic accidents (Bhattachan et al., 2019). Street watering constitutes one of the most common and essential services provided by municipal departments (Gambatese et al., 2001). In such a service, sprinkler trucks, also known as water carts or water trucks, are assembled into a fleet to spray water and wash the road surface alongside streets in urban networks. As a crucial component in street watering operation systems, sprinkler truck route design aims to determine a set of optimal routes for a fleet of sprinkler trucks such that total cost is minimized while road cleaning tasks can be completed as required.

In this study, we use the SRP as an example to illustrate the modeling framework and solution approaches for RARPs on urban networks. The SRP studied in this work is essentially a capacitated arc routing problem with time window (CARPTW), with the consideration of following additional rich features:

- (1) time-varying traffic conditions on urban transportation networks,
- (2) turn delays at intersections,

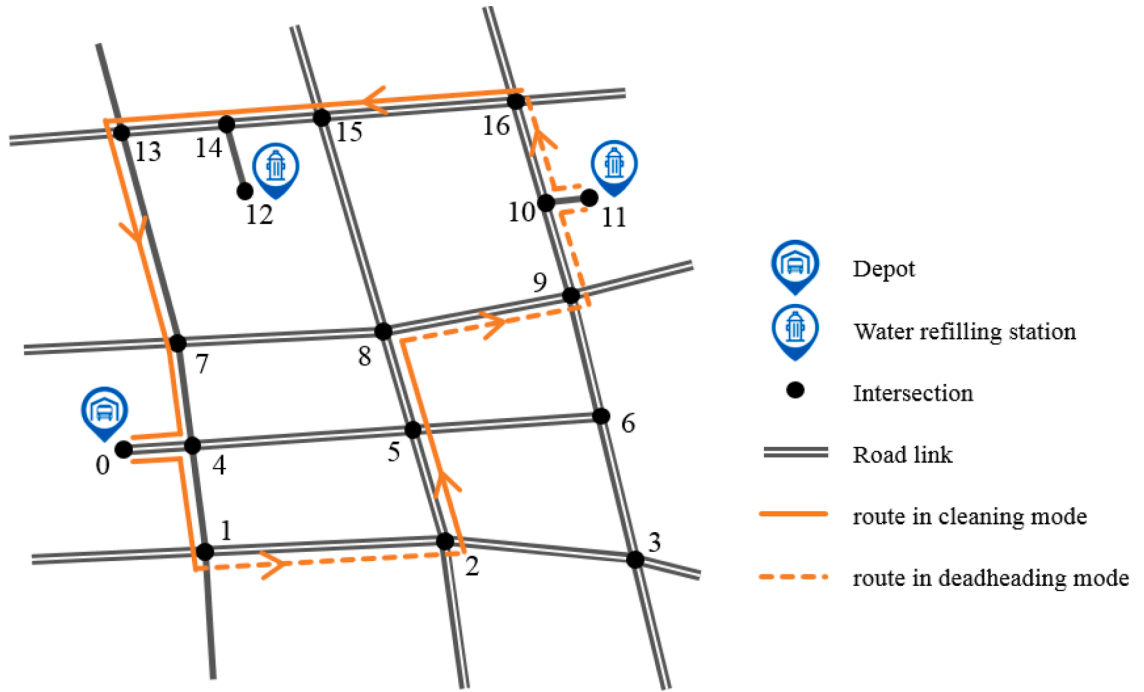


Fig. 4. An illustrative network and the route of a sprinkler truck.

- (3) repeated cleaning services on certain links, and
- (4) water refilling at water refilling stations.

It should be noted that the rich constraints listed above can also be generalized to many other city logistics applications. For example, for the emerging electric vehicle routing problem in green logistics, the rich constraint (4) can be changed to charging vehicles at charging stations, without changing the essence of the resulting problem.

For a given transportation network $G = (N, L)$, where N and L denote the set of nodes and road links respectively, the SRP studied in this paper is to find a set of routes for sprinkler trucks such that all cleaning tasks can be fulfilled as required and the total cost is minimized. Each link $(i, j) \in L$ is associated with a cleaning task. That is, link (i, j) must be cleaned $m_{i,j} \in \mathbb{Z}$ times within its time window $[s_{i,j}, e_{i,j}]$, where $s_{i,j}$ and $e_{i,j}$ denote the earliest and latest service starting time, respectively. To clean link (i, j) once, a sprinkler truck will consume $w_{i,j}$ unit of water. Some nodes belonging to N also serve as water refilling stations. For simplicity, in this paper, we consider the following three assumptions on water consumption and refilling: (a) once a sprinkler truck starts to clean a road link, it must clean the whole link. In other words, cleaning part of a link then going to refill water is not allowed; (b) a sprinkler truck is always refilled to its maximum water tank capacity when visiting a water refilling station, and the time used for refilling is fixed, e.g., 5 minutes, no matter how much water left before visiting a water refilling station; (c) sprinkler trucks are full of water when departing from their origin depot. Due to the existence of assumption (a), the water consumption $w_{i,j}$ of each link (i, j) should not exceed the maximum water capacity C of sprinkler trucks. In the case that $w_{i,j}$ is larger than C , link (i, j) will be split into multiple short links, with each of which meeting the requirements mentioned above.

The total cost to be optimized is calculated by $TOC + \omega \times TSIC$, where TOC and $TSIC$ represent total operating cost and total system-wide impact cost respectively; parameter ω is a user-defined weight of $TSIC$ to measure the importance of societal cost in routing solutions. The societal impact of using a specific link has been derived in Eq. (14). The total operation cost TOC consists of sprinkler truck acquisition cost and total travel time cost. We assume all sprinkler trucks used are identical, and acquisition cost h will be applied for each used sprinkler truck. Besides, the number of available sprinkler trucks is unlimited. The travel time cost of each sprinkler truck equals to the time difference between leaving depot and returning back to the depot, which further consists of four parts: cleaning time, deadheading time, waiting time, and water refilling time. Deadheading means a sprinkler truck traverses a road link without cleaning the link. It may occur in two situations: (a) the link does not need to clean or has been cleaned; (b) a sprinkler has used up water and is heading to a water refilling station. For each road link, its time-dependent deadheading time $DT(t)$ and cleaning time $CT(t)$ are calculated as follows:

$$DT(t) = w(t) + LL/v_d, \quad (15)$$

$$CT(t) = \max[DT(t), LL/v_c], \quad (16)$$

where $w(t)$ is time-dependent link delay derived in Eq. (7); LL represent the length of the link, v_d and v_c denote the maximum speed of

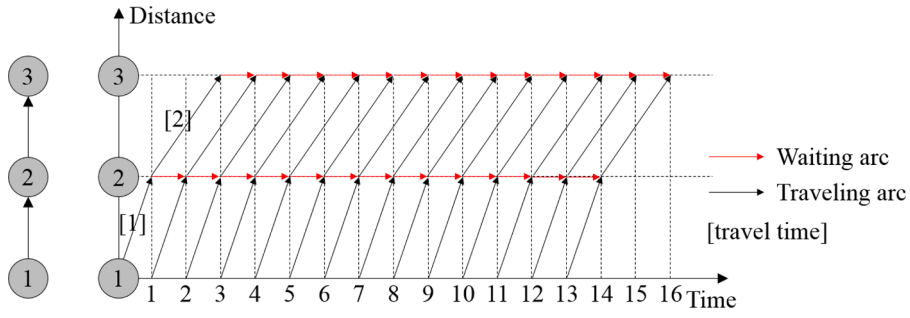
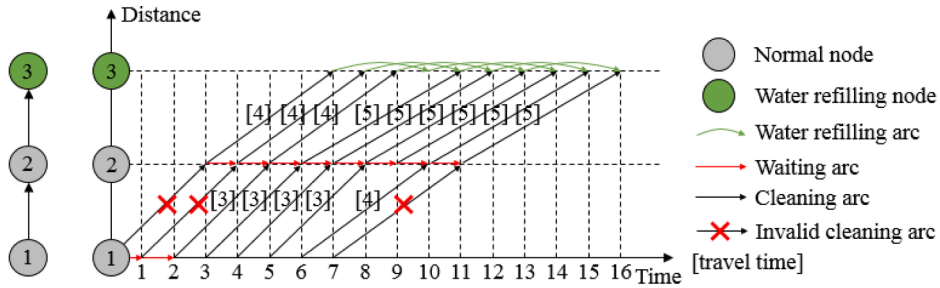
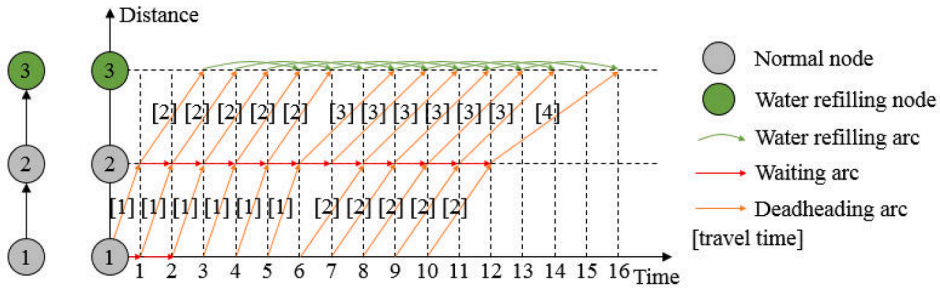


Fig. 5. A simple network and its corresponding time-expanded network.



(a) Physical network and its corresponding time-expanded network (with cleaning arcs)



(b) Physical network and its corresponding time-expanded network (with deadheading arcs)

Fig. 6. A simple network and its corresponding time-expanded networks in SRP.

sprinkler trucks in the deadheading mode and cleaning mode respectively. Waiting time represents the time gap between the arrival time and service starting time of a sprinkler on links (due to time window), rather than the waiting time at intersections caused by control delay. Note that we do not allow sprinkler trucks to wait if they do not have to, e.g., keep staying at a link after cleaning service is complete.

Fig. 4 shows an illustrative example with a simple network and a route of a sprinkler truck. The network consists of 17 nodes (intersections) indexed from 0 to 16. The depot is located at node 0, and two water refilling stations are located at node 11 and 12 respectively. The orange line denotes an example route, which is composed of solid lines (truck in cleaning mode) and dash lines (truck in deadheading mode). In Fig. 4, the sprinkler truck starts from depot node 0, cleans links (0,4), (4,1), (2,5), (5,8), refills water at node 11, cleans links (16,15), (15,14), (14,13), (13,7), (7,4), (4,0), and returns back to depot node 0.

4. Mathematical formulations of the optimization problem

In this section, three models are developed for the proposed SRP from different perspectives. Specifically, a discretized time-expanded network based arc routing model M1 in Section 4.1, an arc-based node routing model M2 in Section 4.2, and a path-based node routing model M3 in Section 4.3. Section 4.4 offers a comprehensive comparison.

Table 2
Notations used in model M1-TEN.

Symbols	Definition
Indices	
i, j	Index of nodes in graph $G_{\mathcal{J}}$
t, t'	Index of time intervals
f	Index of sprinkler trucks
Sets	
$L_{\mathcal{J}}$	Set of links in graph $G_{\mathcal{J}}$
F	Set of sprinkler trucks
V	Set of vertexes in graph $G_{\mathcal{J}\mathcal{T}}$
A	Set of arcs in graph $G_{\mathcal{J}\mathcal{T}}$
CA_{ij}	Cleaning arc set associated with link $(i, j) \in L_{\mathcal{J}}$
RA	Set of water refilling arcs in graph $G_{\mathcal{J}\mathcal{T}}$
Parameters	
$o_{\mathcal{J}\mathcal{T}}(d_{\mathcal{J}\mathcal{T}})$	Origin (destination) vertex of sprinkler trucks in graph $G_{\mathcal{J}\mathcal{T}}$
$ct_{ij, t, t'}$	Travel time cost of arc $(i, j, t, t') \in A$
$cg_{ij, t, t'}$	System impact cost of arc $(i, j, t, t') \in A$
$c_{ij, t, t'}$	General cost of arc $(i, j, t, t') \in A$
m_{ij}	Number of cleaning requests of link (i, j) in graph $G_{\mathcal{J}}$
$w_{ij, t, t'}$	Water consumption of arc $(i, j, t, t') \in A$
C	Water tank capacity of sprinkler trucks
r	Water refilling time
h	Sprinkler truck acquisition cost
Variables	
$x_{ij, t, t'}^f$	Binary variable. $x_{ij, t, t'}^f = 1$ if sprinkler f uses arc (i, j, t, t') ; otherwise, $x_{ij, t, t'}^f = 0$
q_{it}^f	Water level of sprinkler f at vertex (i, t)

4.1. An arc routing model based on time-expanded network (model M1-TEN)

A standard VRP or ARP typically assumes, each customer must be served once and exactly once. Benefitting from the assumption, vehicle states (arrival time, cumulative loads) can be associated with customers, thus a concise physical network based model can be built (Cordeau, 2006). However, in the SRP considered in this study, some road links may be required to be cleaned multiple times, implying a road link may be serviced multiple times by sprinkler trucks, which makes vehicle states intractable if associating them with physical networks. As a result, we adopt a time-expanded network based modeling approach. In the literature, the time-expanded network based modeling approach has been successfully applied in solving a wide range of transportation supply-side optimization problems, e.g., dynamic traffic assignment (Lu et al., 2016), VRP (Yao et al., 2019), passenger flow state estimation (Shang et al., 2019), and train timetabling problem (Zhang et al., 2019). By extending a physical network to a time-expanded network, nodes and road links in the original physical network are extended to vertexes and arcs with an extra time dimension. With the extended time dimension, the multiple-services requirements can be systematically modelled as well as time-dependent travel times. To enable model M1 to accommodate the need of turn delay modeling, with the intersection expansion process similar to the approach discussed by Kirby and Potts (1969), Ziliaskopoulos and Mahmassani (1996), and Pallottino and Scutella (1998), a new network $G_{\mathcal{J}} = (N_{\mathcal{J}}, L_{\mathcal{J}})$ is first constructed from the network G described in Section 3.

For a time-expanded network $G_{\mathcal{J}\mathcal{T}}$ from $G_{\mathcal{J}}$, vertex $(i, t) \in V$ is constructed from physical node $i \in N_{\mathcal{J}}$, where t denotes time. Arc $(i, j, t, t') \in A$ represents a space-time traveling activity from vertex (i, t) to vertex (j, t') . As the time dimension is continuous, we evenly discretize the entire planning horizon $[0, T]$ into short time intervals, e.g., 10 seconds, so that the number of vertexes and arcs is finite. As a result, t and t' both represent the indices of discretized time intervals. We use $\chi = \{0, 1, 2, \dots, s\}$ to denote the set of indices of discretized time intervals, where s is the index of the last time interval. Arcs in a time-expanded network consist of two categories: traveling arc (i, j, t, t') and waiting arc $(i, i, t, t + 1)$. A traveling arc (i, j, t, t') means a vehicle enters link $(i, j) \in L_{\mathcal{J}}$ at time t and leaves at time t' , and $t' - t$ equals the travel time of the link at time t . Note that link (i, j) can either be a road link or a movement link in network $G_{\mathcal{J}}$. A waiting arc $(i, i, t, t + 1)$ corresponds to the waiting activity of a vehicle at node $i \in N_{\mathcal{J}}$ for one time interval. Waiting arcs are used when a sprinkler truck arrives at a road link earlier than the link's earliest service starting time. Fig. 5 depicts a simple three-node network with its corresponding time-expanded network, where the travel time of link (1, 2) and link (2, 3) are 1 and 2, respectively.

With the basic concepts introduced above, we present the time-expanded network built for the SRP in Fig. 6, with the following problem-specific highlights:

- (1) **Cleaning arc and deadheading arc:** When a sprinkler moves on a road link, it has two possible modes, i.e., cleaning mode and deadheading mode. In the cleaning mode, the sprinkler truck cleans the link when traveling on it, but not in the deadheading mode. Note that sprinkler trucks typically have different speeds in the cleaning mode and deadheading mode. Therefore, it is necessary to build cleaning arcs (Fig. 6(a)) and deadheading arcs (Fig. 6(b)) with different travel times for each physical link. For a cleaning arc (i, j, t, t') in Fig. 6(a), $t' - t$ equals to cleaning time on link (i, j) ; for a deadheading arc (i, j, t, t') in Fig. 6(b), $t' - t$ equals to deadheading time on link (i, j) . Another difference between cleaning arcs and deadheading arcs is about water

consumption. If we use $w_{ij,t,t'}$ to denote the water consumption of arc (i, j, t, t') , $w_{ij,t,t'} = w_{ij}$ if the arc is a cleaning arc; otherwise, $w_{ij,t,t'} = 0$.

- (2) **Time-dependent travel time:** For a deadheading arc (i, j, t, t') , $t' - t$ equals to $DT(t)$ in Eq. (15); for a cleaning arc (i, j, t, t') , $t' - t$ equals to $CT(t)$ in Eq. (16). One can observe that each traveling arc (including deadheading arc and cleaning arc) in Fig. 6 is associated with its own travel time, which can be viewed as a fine discretized approximation of continuous functions in Eqs. (15) and (16).
- (3) **Service time window:** In the SRP, each link is associated with a service time window, within which cleaning services must be started. In other words, for a specific link, cleaning arcs outside its service time window are not allowed to use. We can easily impose this constraint by deleting cleaning arcs outside service time windows when building a time-expanded network. For example, in Fig. 6(a), the service time window of link $(1, 2)$ is $[2, 6]$, then the arcs with a red cross will be deleted.
- (4) **Waiting arc:** According to the problem description in Section 3, waiting is only allowed when a sprinkler truck arrives earlier than link service starting time. Therefore, compared with the illustrative example in Fig. 5, tighter restrictions are considered in building waiting arcs for the SRP to avoid invalid waiting. First, waiting arcs are only built for inbound nodes of road links that need cleaning services. Second, waiting arcs with entering time later than the corresponding link's service starting time will not be generated. For road link $(1, 2)$ with time window $[2, 6]$ in Fig. 6, waiting arcs $(1, 1, t, t+1)$ are only constructed for $t = 0, 1$.
- (5) **Water refilling arc:** Water refilling arcs $(i, i, t, t+r)$ are built on each water refilling station i , where r denotes the time required to refill a sprinkler truck.
- (6) **Origin and destination:** In the newly generated time-expanded network, vertex $o_{\mathcal{T}}(o, 0)$ and $d_{\mathcal{T}}(d, s)$ will serve as the origin and destination vertex of sprinkler trucks, where o and d denote the origin node and destination node in network G ; s is the index of the last discretized time interval in the whole planning horizon. Waiting arcs on destination d are constructed to ensure that sprinkler trucks are able to return back to and keep staying at the destination depot after finishing cleaning tasks.

To reduce the size of time-expanded networks and simplify the subsequent optimization model, we can further remove vertexes and arcs outside the space-time prism between the origin vertex $o_{\mathcal{T}}$ and destination vertex $d_{\mathcal{T}}$. A space-time prism is an envelope that covers all possible paths between two space-time vertexes. Interested readers are referred to Miller (2005) and Tong et al. (2015).

With the preparation of intersection expansion and time-expanded network construction, we now present the arc routing formulation on time-expanded network (model M1-TEN) for the proposed SRP. Notations are summarized in Table 2.

Model M1-TEN:

Objective function

$$\min Z_1 = \sum_{f \in F} \sum_{(i,j,t,t') \in A} ct_{ij,t,t'} x_{ij,t,t'}^f + h \sum_{f \in F} \sum_{(j,i,t,t') \in A} x_{oj,0,t'}^f + w \sum_{f \in F} \sum_{(i,j,t,t') \in A} cg_{ij,t,t'} x_{ij,t,t'}^f \quad (17)$$

Subject to:

Flow balance constraint:

$$\sum_{(j,i,t,t') \in A} x_{ij,t,t'}^f = \sum_{(j,i,t,t') \in A} x_{ji,t,t'}^f, \forall f \in F, (i, t) \in V \setminus \{o_{\mathcal{T}}, d_{\mathcal{T}}\} \quad (18)$$

Cleaning request satisfaction constraint:

$$\sum_{f \in F} \sum_{(i,j,t,t') \in CA_{ij}} x_{ij,t,t'}^f = m_{ij}, \forall (i, j) \in L_{\mathcal{T}} \quad (19)$$

Sprinkler truck water level updating constraint:

$$q_{j,t'}^f \leq q_{i,t}^f - w_{ij,t,t'} x_{ij,t,t'}^f + C(1 - x_{ij,t,t'}^f), \forall f \in F, (i, j, t, t') \in A/RA \quad (20)$$

Decision variables:

$$x_{ij,t,t'}^f \in \{0, 1\}, \forall f \in F, (i, j, t, t') \in A$$

$$0 \leq q_{i,t}^f \leq C, \forall f \in F, (i, t) \in V \quad (21)$$

The objective function in Eq. (17) minimizes the total cost to complete cleaning tasks, which includes three parts: travel time cost $\sum_{f \in F} \sum_{(i,j,t,t') \in A} ct_{ij,t,t'} x_{ij,t,t'}^f$, vehicle acquisition cost $h \sum_{f \in F} \sum_{(j,i,t,t') \in A} x_{oj,0,t'}^f$, and system impact cost $w \sum_{f \in F} \sum_{(i,j,t,t') \in A} cg_{ij,t,t'} x_{ij,t,t'}^f$. The travel time cost $ct_{ij,t,t'}$ of arc (i, j, t, t') equals to $t' - t$ for all arcs in set A , except waiting arcs built at sprinkler destination node d . We let the cost of waiting arcs at node d equal to 0 (i.e., $c_{d,i,t,t+1} = 0$ if $i = d$). Expression $\sum_{f \in F} \sum_{(j,i,t,t') \in A} x_{ij,t,t'}^f$ denotes the number of sprinkler trucks used. By integrating three types of coefficient of each $x_{ij,t,t'}^f$, Eq. (17) can be simplified as

$$\min Z_1 = \sum_{f \in F} \sum_{(i,j,t,t') \in A} c_{ij,t,t'} x_{ij,t,t'}^f, \quad (22)$$

Table 3
Additional notations used in the node routing model.

Symbols	Definition
Indices	
i, j	Index of nodes in graph $G_{\mathcal{F}}$
ρ	Index of edges between each pair of nodes in graph $G_{\mathcal{F}}$
p	Index of time slots
Sets	
$N_{\mathcal{F}}$	Set of nodes in graph $G_{\mathcal{F}}$
$N_{\mathcal{F}}^s$	Set of service nodes in graph $G_{\mathcal{F}}$
$N_{\mathcal{F}}^{wv}$	Set of water refilling stations in graph $G_{\mathcal{F}}$
$E_{\mathcal{F}}$	Set of edges in graph $G_{\mathcal{F}}$
$P_{\mathcal{F}}$	Set of time slots
Parameters	
ϕ	Number of edges between each pair of nodes in graph $G_{\mathcal{F}}$
$\tau_{i,p}$	Service time of node $i \in N_{\mathcal{F}}$ in slot p
$\tau_{i,j,\rho,p}$	Travel time of edge $(i,j,\rho) \in E_{\mathcal{F}}$ in slot p
$cg_{i,p}$	Congestion impact of serving node $i \in N_{\mathcal{F}}$ in slot p
$cg_{i,j,\rho,p}$	Congestion impact of using edge $(i,j,\rho) \in E_{\mathcal{F}}$ in slot p
w_i	Water consumption of node $i \in N_{\mathcal{F}}$
$s_i(e_i)$	The earliest (latest) service starting time of node $i \in N_{\mathcal{F}}$
$T_{i,p}$	Ending time of slot p for node i
Variables	
$x_{i,j,\rho,p}$	Binary variable. $x_{i,j,\rho,p} = 1$ if a sprinkler uses edge (i,j,ρ) in slot p ; otherwise, $x_{i,j,\rho,p} = 0$
$y_{i,p}$	Binary variable. $y_{i,p} = 1$ if cleaning service starts on node i in slot p ; otherwise, $y_{i,p} = 0$
q_i	Water level of a sprinkler at node i
t_i^d	Departure time of a sprinkler from node i
t_i^s	Service starting time of node i

where $c_{i,j,t,t'}$ represents the general cost of arc (i,j,t,t') , consisting of travel time cost, vehicle acquisition cost, and system impact cost. Constraint (18) guarantees that the incoming flow equals to outgoing flow on vertexes. Note that we do not impose this constraint on the origin vertex $o_{\mathcal{F}}$ and destination vertex $d_{\mathcal{F}}$. Constraint (19) makes sure that links are cleaned as requested, while constraint (20) is used to update the water level of sprinkler trucks. Finally, constraint (21) specifies decision variables with their domains. Specifically, $x_{i,j,t,t'}^f$ are binary variables, and $q_{i,t}^f$ are positive continuous variables with an upper bound of C , where C represents the water tank capacity of sprinkler trucks.

4.2. Graph transformation and a slot-based time discretization node routing model (model M2-STD)

The adoption of intersection expansion and time-expanded networks can allow the consideration of various rich constraints into the generated network, contributing to a concise form of model M1-TEN. Yet, additional movement links created at intersections and high-dimension time-expanded networks significantly increase the size of model M1-TEN, making it extremely challenging to solve on large-scale instances. In this section, we further propose a node routing model by converting the original ARP to a node routing problem (NRP). Solving an ARP by converting it to an NRP was first proposed by Pearn et al. (1987) and was adopted and improved by Longo et al. (2006). The core idea behind the conversion process is treating arcs to be served as activity nodes in the NRP and building virtual edges to connect each pair of the resulting nodes based on the shortest paths in the original network.

In the proposed SRP, due to the consideration of additional rich features, there are three major challenges during the conversion process. First, on a congested urban transportation network with time-varying traffic conditions, the shortest path (in terms of travel time) between two nodes changes during the day. Second, cost of a path consists of travel time cost and system impact cost, therefore a shortest path with the least travel time between two activity nodes does not necessarily correspond to the best path with the least total cost. Third, due to the existence of service time window, even the best path with the least total cost does not guarantee an optimal solution. Consider two paths (e.g., path 1 and path 2) between a pair of activity nodes, where path 1 has a lower total cost than path 2. However, path 1 takes longer travel time, causing some future service nodes not to be served from path 1 due to time windows. Adopting path 1 with a lower total cost may need additional vehicles to visit these unserved service nodes, which could result in a suboptimal routing solution. To address this issue, we keep ϕ paths between each pair of activity nodes in the resulting NRP. The details of the transformation process are illustrated in Appendix A.

Due to the adoption of nonlinear functions for representing time-varying traffic conditions in the converted graph $G_{\mathcal{F}}$, i.e., $\tau_i(t)$, $\tau_{i,j,\rho}(t)$, and $cg_{i,j,\rho}(t)$, one can expect that an optimization model built on $G_{\mathcal{F}}$ is highly nonconvex and extremely hard to solve (see Appendix B). Therefore, in model M2-STD, nonlinear functions are approximated by piecewise-constant functions. That is, the entire planning time horizon $[0, T]$ is split into multiple time slots with the same duration, e.g., 10 minutes, and nonlinear functions are approximated by constants within each slot. Model M2-STD is presented as follows, where notations are illustrated in Table 3.

Model M2-STD:

Objective function

$$\min Z_2 = \sum_{f \in F} t_{d_{f'}}^d + h \sum_{(j,\rho):(o_{f'},j,\rho,p) \in E_{f'}, p \in P_{f'}} x_{o_{f'},j,\rho,p} + \omega \left(\sum_{(i,j,\rho) \in E_{f'}, p \in P_{f'}} c g_{i,j,\rho,p} x_{i,j,\rho,p} + \sum_{i \in N_{f'}, p \in P_{f'}} c g_{i,p} y_{i,p} \right) \quad (23)$$

Subject to:

Cleaning request satisfaction constraint:

$$\sum_{(j,\rho):(j,i,\rho) \in E_{f'}, p \in P_{f'}} x_{j,i,\rho,p} = 1, \quad \forall i \in N_{f'}^s \quad (24)$$

Sprinkler truck spatial route constraint:

$$\sum_{(j,\rho):(j,i,\rho) \in E_{f'}, p \in P_{f'}} x_{j,i,\rho,p} = \sum_{(j,\rho):(i,j,\rho) \in E_{f'}, p \in P_{f'}} x_{i,j,\rho,p}, \quad \forall i \in N_{f'}^s \cup N_{f'}^w \quad (25)$$

$$\sum_{(j,\rho):(j,i,\rho) \in E_{f'}, p \in P_{f'}} x_{j,i,\rho,p} \leq 1, \quad \forall i \in N_{f'}^w \cup \{d_{f'}^f | f \in F\} \quad (26)$$

Sprinkler truck temporal route constraint:

$$t_{o_{f'}}^d = 0 \quad (27)$$

$$t_j^s \geq t_i^d + \tau_{i,j,\rho,p} + M(x_{i,j,\rho,p} - 1), \quad \forall (i,j,\rho) \in E_{f'}, \forall p \in P_{f'} \quad (28)$$

$$t_i^d \geq t_i^s + \tau_{i,p} y_{i,p}, \quad \forall i \in N_{f'}, \forall p \in P_{f'} \quad (29)$$

$$\sum_{p \in P_{f'}} y_{i,p} = 1, \quad \forall i \in N_{f'} \quad (30)$$

$$T_{p-1} x_{i,j,\rho,p} \leq t_i^d < T_p + M(1 - x_{i,j,\rho,p}), \quad \forall (i,j,\rho) \in E_{f'}, \forall p \in P_{f'} \quad (31)$$

$$T_{p-1} y_{i,p} \leq t_i^s < T_p + M(1 - y_{i,p}), \quad \forall i \in N_{f'}, \forall p \in P_{f'} \quad (32)$$

Time window constraint:

$$s_i \leq t_i^s \leq e_i, \quad \forall i \in N_{f'} \quad (33)$$

Sprinkler truck water level updating constraint:

$$q_{o_{f'}} = C \quad (34)$$

$$q_j \leq q_i - x_{i,j,\rho,p} w_j + C(1 - x_{i,j,\rho,p}), \quad \forall (i,j,\rho) \in \{E_{f'} | i \in N_{f'}^s\}, \forall p \in P_{f'} \quad (35)$$

$$q_j \leq C - x_{i,j,\rho,p} w_j, \quad \forall (i,j,\rho) \in \{E_{f'} | i \in N_{f'}^w \cup \{o_{f'}\}\}, \forall p \in P_{f'} \quad (36)$$

Decision variables:

$$\begin{aligned} x_{i,j,\rho,p} &\in \{0, 1\}, \quad \forall (i,j,\rho) \in E_{f'}, \forall p \in P_{f'} \\ y_{i,p} &\in \{0, 1\}, \quad \forall i \in N_{f'}, \forall p \in P_{f'} \\ 0 &\leq q_i \leq C, \quad t_i^d \geq 0, \quad t_i^s \geq 0, \quad \forall i \in N_{f'} \end{aligned} \quad (37)$$

The objective function in Eq. (23) aims to minimize the total cost, where $o_{f'}$ represents the origin node of sprinkler trucks in graph $G_{f'}$, and $d_{f'}^f$ denotes the copy of destination node $d_{f'}$ for sprinkler f in graph $G_{f'}$. By creating a dummy copy $d_{f'}^f$ of the destination depot for each sprinkler truck f , the arrival time of sprinkler truck f at the destination depot $d_{f'}^f$ is the time used by sprinkler truck f . Note that since service time at destination depots is zero, arrival times and departure times at destination depots are the same, thus we can use departure time $t_{d_{f'}^f}^d$ in the objective function to avoid defining extra variables. Constraint (24) ensures that each service node is serviced exactly once. Constraint (25), (26) and Eqs. (27)–(32) formulate the spatial and temporal route constraints of sprinkler trucks, respectively. In constraint (29), $\tau_{i,p}$ denotes the service time of node i if service starts in time slot p . For service nodes, $\tau_{i,p}$ equals to the cleaning time of corresponding road links in slot p . For water refilling stations and depot nodes, $\tau_{i,p}$ equals to the water refilling time and zero, respectively. Constraint (33) makes sure that cleaning services start within preset time windows. Constraints (34)–(36) are used to track the water level updating of sprinkler trucks. Finally, constraint (37) specifies decision variables and their domains used in model M2-STD.

In addition to the essential constraints (24)–(37), two sets of constraints are further constructed below to tighten the linear relaxation of model M2-STD. Specifically, constraint (38) ensures that at least one sprinkler is needed to complete cleaning tasks. Note

Table 4
Parameters of polynomial travel time functions.

Link	t_0	t_3	γ	μ	$FFTT$	Link travel time function
1-2	3	16	15	1200	1.36	$LT(t) = \frac{15}{3 \times 1200}(t-3)^2(16-t) + 1.36$
2-3	4	18	12	1000	1.54	$LT(t) = \frac{12}{3 \times 1000}(t-4)^2(18-t) + 1.54$

that, as sprinkler trucks are allowed to refill water during their trips, we cannot calculate the minimum number of sprinkler trucks using total water needed divided by water tank capacity of sprinkler trucks. Although constraint (38) seems to be very loose, it does serve as the lower bound of many instances in the numerical experiment section. Constraint (39) is used to break the symmetry of model M2-STD.

$$\sum_{(j,p):(o,r,j,p) \in E_{j,r}, p \in P_{j,r}} x_{o,r,j,p} \geq 1 \quad (38)$$

$$t_{d_{j,r}}^d \leq t_{d_{j,r}}^d \leq \dots \leq t_{d_{j,r}^{[F]-1}}^d \leq t_{d_{j,r}^{[F]}}^d \quad (39)$$

4.3. Path-based set partition formulation with continuous-time representation (model M3-CTR)

In this section, we further propose a path-based model (set partition formulation) on the graph $G_{j,r}$ constructed in Appendix A. Instead of time discretization, the original continuous polynomial form of service time function and travel time function is adopted. As pointed by Boland et al. (2017), the granularity of the discretization has an impact on both candidate solutions and the computational tractability. Let $r \in \Omega$ denote a feasible sprinkler truck route, where Ω represents the set of all feasible routes. A feasible route r should satisfy constraints (B3–B12) in Appendix B. The set partition formulation can be expressed as follows:

Model M3-CTR:

Objective function

$$\min Z_3 = \sum_{r \in \Omega} c_r \theta_r \quad (40)$$

Subject to:

Cleaning request satisfaction constraint:

$$\sum_{r \in \Omega} \alpha_{i,r} \theta_r = 1, \quad \forall i \in N_{j,r}^s \quad (41)$$

Decision variables:

$$\theta_r \in \{0, 1\}, \quad \forall r \in \Omega \quad (42)$$

The objective function Eq. (40) minimizes the cost of all selected routes, where θ_r is a binary variable indicating whether route r is chosen in a solution or not; c_r is the cost of route r , including operation cost and system impact. Constraint (41) states that all service nodes are serviced once, where $\alpha_{i,r}$ is a mapping coefficient between service node i and route r (number of times that route r passes activity node i).

4.4. Model comparison in terms of time-varying travel time representation and model formulation

An illustrative example with two consecutive links is first used to show the difference in time-dependent travel time modeling among the three models. Table 4 presents the sample values for the parameters of polynomial travel time functions proposed in Section 2. The time unit in this example is minute.

Fig. 7 shows the travel time functions (blue lines) used in the three models, among which piecewise-constant functions depicted in Fig. 7(b) correspond to the first method in Table 1, while functions in Fig. 7(a) and Fig. 7(c) are the two proposed approaches in this study. For model M1-TEN, the time dimension is discretized into 0.2-minute time intervals. Therefore, both link travel times and node arrival (departure) times are approximated with a 0.2-minute resolution. For model M2-STD, the entire time horizon is split into four 5-minute slots, and travel times remain constant in each slot. For model M3-CTR, the original polynomial form of travel time functions is used.

In Table 5, a comprehensive summary of the three proposed models is further provided. For the formulation with continuous-time representation M3-CTR, it is important to use a well-behaved travel time function satisfying the FIFO property; otherwise, unnecessary and unrealistic waiting could be observed from the final solutions. In Section 6, extensive numerical experiments will be performed to examine the computational efficiency of the three models.

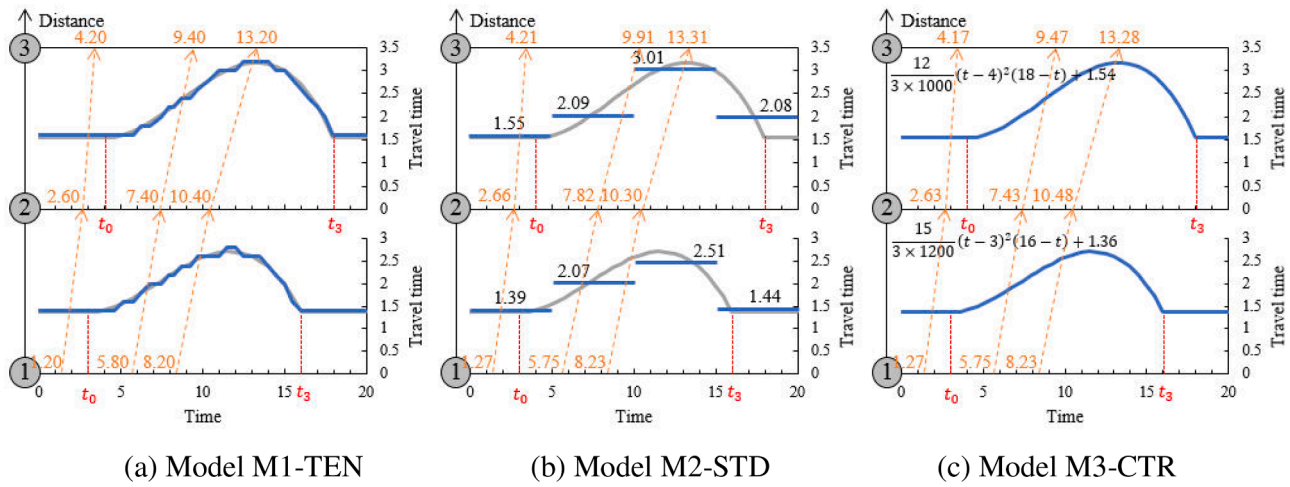


Fig. 7. An illustrative example of time-dependent travel time modeling in three models, ranging from high-fidelity discretization M1-TEN, semi-dynamic slot-based discretization M2-STD and continuous-time representation M3-CTR.

Table 5
Comparisons among the three proposed models.

	Model M1-TEN	Model M2-STD	Model M3-CTR
Model type	Arc routing model (MILP)	Arc-based node routing model (MILP)	Path-based node routing model (MILP)
Network type	Time-expanded networks from physical networks	Activity node-based networks from physical networks	
Rich constraint handling	(1) Embedded in arcs; approximated with a fine resolution TV	Piecewise-constant approximation; extra slot dimension	Directly using polynomial travel time functions for an entire congestion period
	(2) Intersection expansion	Embedded in edges between activity nodes	
	(3) Allow multiple visits along extra time dimension	Dummy activity nodes	Dynamically considered when constructing a path
	(4) Water refilling arcs	Dummy water refilling nodes	
	WR		
Main decision variables	$x_{ij,t,t'}^f \in \{0, 1\}, \forall f \in F, (i, j, t, t') \in A$	$x_{i,j,p} \in \{0, 1\}, \forall (i,j) \in E_{\mathcal{A}}, \forall p \in P_{\mathcal{A}}$ $y_{i,p} \in \{0, 1\}, \forall i \in N_{\mathcal{A}}, \forall p \in P_{\mathcal{A}}$	$\theta_r \in \{0, 1\}, \forall r \in \Omega$
Side constraints	Flow balance constraint; Cleaning request satisfaction constraint; Sprinkler truck water level updating constraint	Sprinkler spatial route constraint; Sprinkler temporal route constraint; Cleaning request satisfaction constraint; Time window constraint; Sprinkler truck water level updating constraint	Cleaning request satisfaction constraint
Features	Embed various rich constraints into time-expanded networks, resulting in a concise model formulation; Fine time-dependent travel time modeling	Turn delays are considered in the activity node-based graph construction stage, without increasing the complexity of subsequent optimization models Less number of decision variables compared with the other two models	Only one set of side constraint
Challenges	A large number of additional decision variables are introduced due to intersection expansion and the use of time-expanded network	Time-dependent travel times are approximated with piecewise-constant functions and modeled with an additional slot dimension which increases the model size (complexity)	For large-scale instances, feasible paths cannot be explicitly numerated.

Notes: MILP – mixed integer linear programming, TV - time-varying traffic conditions on urban transportation networks, TD - turning delays at intersections, RS - repeated cleaning services on certain links, WR - water refilling at water refilling stations.

5. Exact solution methods

The models developed in Section 4 are all (mixed) integrated linear programming models and theoretically solvable using existing solvers such as Gurobi and CPLEX. While, in real-life applications, the size of resulting models could be extremely large, making it hard to obtain desirable solutions from solvers within a reasonable time. In this section, recognizing the mathematical characteristics of each proposed model, we develop two exact solution methods for efficiently solving these models. Specifically, a Lagrangian relaxation (LR) based method is proposed for solving model M1-TEN, while a branch-and-price (BnP) based approach is proposed for solving model M3-CTR.

5.1. Lagrangian relaxation

The concise form of model M1-TEN provides the possibility of using decomposition techniques to obtain good solutions quickly. Model M1-TEN consists of three sets of side constraints, among which only cleaning request satisfaction constraint (19) couples different sprinkler trucks. If constraint (19) is relaxed, model M1 can be decomposed into multiple easier sprinkler-specific sub-problems, which can be solved independently. In this work, we propose a LR-based method for solving model M1-TEN.

Under the LR framework, coupling constraint (19) is relaxed, and the violation is penalized in objective function Eq. (17) using Lagrangian multipliers λ_{ij} . The relaxed problem is

Objective function

$$\min Z_4 = Z_1 + \sum_{(i,j) \in L_{\mathcal{S}}} \lambda_{ij} \left(\sum_{f \in F} \sum_{(i,j,t,t') \in CA_{ij}} x_{ij,t,t'}^f - m_{ij} \right) \quad (43)$$

Subject to:

(18), (20), (21).

By reorganizing the coefficients of decision variable $x_{ij,t,t'}^f$, the objective function Eq. (43) can be further simplified to Eq. (44). Given the value of λ_{ij} , the relaxed problem needs to find the least cost shortest path on the discretized time-expanded network with modified arc cost $\hat{c}_{ij,t,t'}'$ for each sprinkler truck, which can be exactly solved by the dynamic programming approach (Mahmoudi and Zhou, 2016; Yao et al., 2019).

$$\min Z_4 = \sum_{f \in F} \sum_{(i,j,t,t') \in A} \hat{c}_{ij,t,t'}' x_{ij,t,t'}^f - \sum_{(i,j) \in L_4} \lambda_{ij} m_{ij} \quad (44)$$

We adopt the classical sub-gradient method to update the values of Lagrangian multipliers λ_{ij} using Eq. (45) and Eq. (46), where λ_{ij}^k and α^k denote the value of λ_{ij} and step length at iteration k , respectively. It should be noted that, as constraint (19) is an equality constraint, Lagrangian multipliers can be positive, negative, or 0. A positive Lagrangian multiplier means an additional penalty when serving the corresponding arc, while a negative value means an additional bonus. Lagrangian iteration stops when constraint (19) is satisfied on all links, i.e., no Lagrangian multipliers updating in Eq. (45).

$$\lambda_{ij}^{k+1} = \lambda_{ij}^k + \alpha^k \left(\sum_{f \in F} \sum_{(i,j,t,t') \in CA_{ij}} x_{ij,t,t'}^f - m_{ij} \right) \quad (45)$$

$$\alpha^k = 1/(k+1) \quad (46)$$

A detailed illustration of the Lagrangian relaxation based solution process is presented in Appendix C. Interested readers are also referred to Zhang et al. (2019) and Yao et al. (2019) for efficiently solving a model with coupling constraints on time-expanded networks under a dual decomposition framework using alternating direction method of multipliers.

5.2. Branch and price algorithm

For model M3-CTR, the number of feasible routes in set Ω can be extremely large in large-scale problems, making it nearly impossible to enumerate all routes and solve the resulting model using existing solvers. In this section, we further propose a Branch and Price (BnP)-based exact solution approach to solve model M3-CTR. The proposed BnP algorithm consists of two modules, i.e., column generation (CG) and branch-and-bound (BnB). Model M3-CTR is denoted as the master problem (MP), then we need to use CG module to solve the linear master problem (LMP), while the BnB module is needed to obtain integer solutions based on the results from the CG module. The LMP is the same as MP except the domain of decision variables. In LMP, we relax the integer requirement of decision variable θ_r , i.e., $0 \leq \theta_r \leq 1$. Due to the existence of constraint (41), we can further simplify the domain constraint $0 \leq \theta_r \leq 1$ to $\theta_r \geq 0$.

Column generation

The LMP is still hard to solve due to the large size of route set Ω . Therefore, instead of directly solving the LMP, we iteratively solve its corresponding restricted linear master problem (RLMP), whose route set Ω' only contains part of feasible routes and is extended with new routes as needed. The RLMP is presented as follows:

Algorithm 1

Label extension procedure.

Input: label $l = \{\text{node}, R_c, R_p, R_w, \text{prev}, \Phi\}$, node service time function $\tau_i(t)$, edge travel time function $\tau_{ij, \rho}(t)$, system impact functions $cg_i(t)$ and $cg_{ij, \rho}(t)$

Output: new label l' on node j

```

1: label  $l' = \text{null}$ 
2: if  $j \notin \Phi$  and  $R_w - w_j \geq 0$  then
3:   arrival time at node  $j$ :  $at_j = R_t + \tau_i(R_t) + \tau_{ij, \rho}(R_t + \tau_i(R_t))$ 
4:   if  $at_j \leq e_j$  then
5:     the earliest service starting time at node  $j$ :  $R'_t = \max(at_j, s_j)$ 
6:     the earliest time to return back to the depot after visiting node  $j$ :
        $at_{d_N} = R'_t + \tau_j(R'_t) + \min_{\rho} \{\tau_{j, d_N, \rho}(R'_t + \tau_j(R'_t))\}$ 
7:     if  $at_{d_N} \leq e_{d_N}$  then
8:        $R'_t = \begin{cases} R_c + R'_t - R_t + cg_{ij, \rho}(R_t + \tau_i(R_t)) + cg_i(R_t) - \pi_j & \text{if } j \in N_{j'}^S \\ R_c + R'_t - R_t + cg_{ij, \rho}(R_t + \tau_i(R_t)) + cg_i(R_t) & \text{if } j \notin N_{j'}^S \end{cases}$ 
9:        $R'_w = \begin{cases} C & \text{if } j \in N_{j'}^w \\ R_w - w_j & \text{if } j \notin N_{j'}^w \end{cases}$ 
10:       $\text{prev}' = l$ 
11:       $\Phi' = \begin{cases} \Phi \cap \mathcal{T}_j & \text{if } j \in N_{j'}^w \\ \Phi \cap \mathcal{T}_j \cup \{j\} & \text{if } j \notin N_{j'}^w \end{cases}$ 
12:      label  $l' = \{j, R'_c, R'_t, R'_w, \text{prev}', \Phi'\}$ 
13:    return label  $l'$ 

```

Objective function

$$\min Z_5 = \sum_{r \in \Omega'} c_r \theta_r \quad (47)$$

Subject to:

Cleaning request satisfaction constraint:

$$\sum_{r \in \Omega'} \alpha_{i,r} \theta_r = 1, \quad \forall i \in N_{j'}^S \quad (48)$$

Decision variables:

$$\theta_r \geq 0, \quad \forall r \in \Omega' \quad (49)$$

In this paper, the initial routes in set Ω' are generated by assigning one sprinkler for each service node. Note that, when generating initial routes, if a node cannot be serviced due to the violation of time window constraint or water consumption constraint, the original problem is then infeasible. The route set Ω' in RLMP is extended by solving the so-called pricing problem, during which routes with negative reduced cost are added into set Ω' .

(a) Formulation of the pricing problem

The pricing problem is formulated as follows:

Objective function

$$\min Z_5 = h + t_{d,j'}^d + \omega \left(\sum_{(i,j,\rho) \in E_{j'}} cg_{ij,\rho}(t_i^d) x_{i,j,\rho} + \sum_{(i,j,\rho) \in E_{j'}} cg_j(t_j^s) x_{i,j,\rho} \right) - \sum_{(i,j,\rho) \in E_{j'}, j \in N_{j'}^S} x_{i,j,\rho} \pi_j \quad (50)$$

Subject to: constraints B3–B12 in Appendix B.

Objective function Eq. (50) minimizes the reduced cost of a feasible route, where π_j is the reduced cost associated with node j . Essentially, the optimization problem presented above is an elementary shortest path problem with resource constraint (ESPPRC).

(b) Dynamic programming algorithm for the pricing problem

Finding an elementary shortest path with resource constraint in a network is an NP-hard problem. Therefore, finding the exact solution for the aforementioned pricing problem may be extremely time-consuming in large-scale instances. In the literature, to reduce the computational complexity of the pricing problem, following the pioneering work by Christofides et al. (1981), researchers started to develop efficient algorithms for finding non-elementary shortest paths with resource constraint through relaxing the elementary requirement, i.e., allowing visiting a node multiple times (Irnich and Villeneuve, 2006; Baldacci et al., 2011; Martinelli et al., 2014). The use of non-elementary shortest paths will not affect finding the optimal solution of the original problem but will weaken its linear relaxations. In this paper, we adopt the dynamic programming ng-route algorithm proposed by Martinelli et al. (2014) with some modifications. Specifically, in Martinelli et al. (2014), a vehicle only needs to collect goods during its trip, therefore the load of a

Algorithm 2

Dynamic programming ng-route algorithm.

Input: node-based graph $G_{\mathcal{F}}$, ng-set $\mathcal{V}_i$ for each node $i \in N_{\mathcal{F}}$
Output: the best ng-route

```

1: create a root label  $l_0 = \{o_{\mathcal{F}}, h, 0, C, null, \{o_{\mathcal{F}}\}\}$ 
2:  $\mathcal{W} \leftarrow \{l_0\}$ 
3: while  $\mathcal{W} \neq \emptyset$  do
4:   choose a label  $l$  with the minimum service starting time from  $\mathcal{W}$ , and delete  $l$  from  $\mathcal{W}$ 
5:   for  $j \in N_{\mathcal{F}}$  do
6:     for  $\rho := 1, 2, \dots, \varphi$  do
7:       if label  $l$  can be extended to node  $j$  via edge  $\rho$  (feasibility checking in Algorithm 1) then
8:         create a new label  $l'$  on node  $j$  from label  $l$ 
9:         insertLabel  $\leftarrow$  true
10:        for label  $l'' \in \mathcal{W}$  on node  $j$  do
11:          if label  $l'$  dominates  $l''$  then
12:            insertLabel  $\leftarrow$  false
13:            break
14:          else if label  $l'$  dominates  $l''$  then
15:            delete  $l''$ 
16:        if insertLabel then
17:           $\mathcal{W} \leftarrow \mathcal{W} \cup \{l'\}$ 
18:      best label  $l^* \leftarrow$  the label with the least reduced cost on the destination depot node
19:      retrieving the best route from label  $l^*$ 
20:      return the best route

```

vehicle keeps increasing along with its trip. As a result, a two-dimension matrix with vehicle load as one axis can be created, and the dynamic programming process can be performed in the order of vehicle load increasing. However, in the SRP considered in this work, sprinkler trucks are allowed to refill water during their trips, hence the water level changes on sprinkler trucks are not monotonous anymore. Another challenge is that demands are assumed to be discrete in Martinelli et al. (2014), while it is continuous in the proposed SRP, thus creating a matrix with sprinkler water level as an axis is impossible.

We first introduce the three key components of a dynamic programming algorithm, including state (label) definition, extension rule, and dominance rule, followed by the modified ng-route algorithm developed in this paper.

Label definition. Labels used in the pricing problem have the following form: $l = \{node, R_c, R_b, R_w, prev, \Phi\}$, where *node* means current node label l is on; R_c, R_b, R_w denotes the reduced cost, the actual service starting time, and the water level of the sprinkler at current node, respectively; *prev* stands for the previous label of label l ; Φ represents the set of forbidden nodes that are not allowed to extend from label l , which will be introduced in the extension rule below.

Extension rule. The process of extending current label l to node j via edge ρ is presented in Algorithm 1. In Algorithm 1, line 2–4 check if node j is reachable from current node i ; line 7 checks if the destination depot is reachable after visiting node j . Line 12 creates a new label l' on node j from label l via edge ρ . Specifically, the ng-set $\mathcal{V}_j$ of node j is used when updating forbidden set Φ' . For each node i , a ng-set $\mathcal{V}_i$ is predefined which includes a certain number of nearest nodes of i (including itself). A label on node i only includes the intersection of visited nodes and $\mathcal{V}_i$ into its forbidden set Φ , resulting in the possibility of forming cycles. Generally, the larger the ng-sets are, the less likely to contain cycles in routes, and also closer to the original ESPPRC (harder to solve). Water refilling stations will also not be put into forbidden sets as they are always allowed to revisit in the SRP settings.

Dominance rule. For two labels l and l' on the same node, label l' is dominated by label l if the following conditions are satisfied and at least one of them is not equal:

- (i) $l.R_c \leq l'.R_c - (l'.R_t - l.R_t)$
- (ii) $l.R_t \leq l'.R_t$
- (iii) $l.R_w \leq l'.R_w$
- (iv) $l.\Phi \subseteq l'.\Phi$

It should be noted that condition (i) is much tighter than the condition that is commonly used in VRPs with the objective of minimizing total travel distance, i.e., $l.R_c \leq l'.R_c$. A stricter dominance rule means less labels can be dominated, thus there are more labels to be processed (the pricing problem is harder to solve). If label l' is dominated by other labels, label l' can then be safely discarded without affecting the final optimal results.

The dynamic programming ng-route algorithm is presented in Algorithm 2. Algorithm 2 finds the best route with given graph $G_{\mathcal{F}}$ and ng-sets $\mathcal{V}_i$, which may be very time-consuming in large instances. When used under CG, to save computing time, Algorithm 2 can be terminated as soon as a certain number of paths with negative reduced cost have been found.

(c) Acceleration techniques

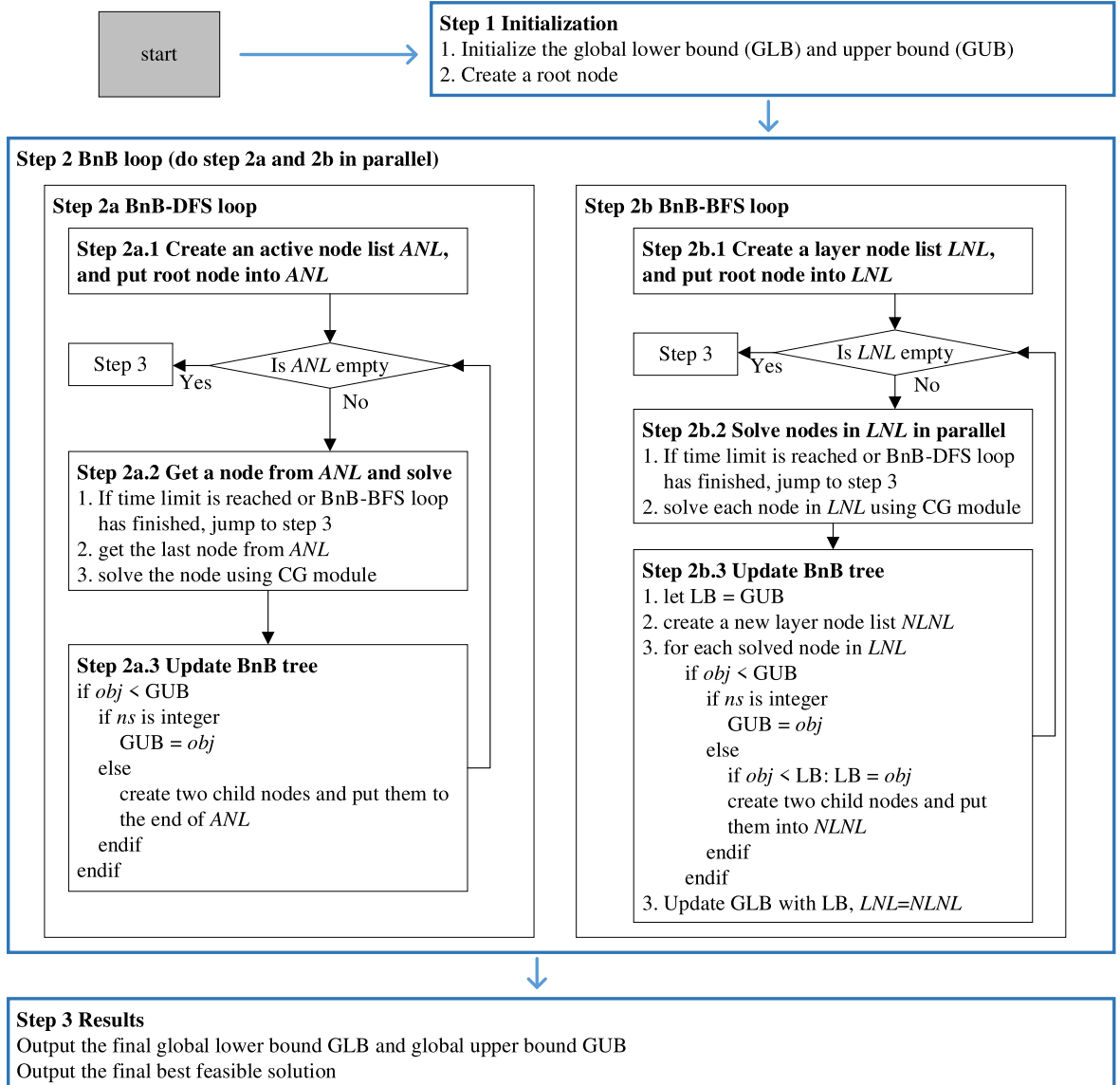
Although finding non-elementary shortest paths using Algorithm 2 is faster than finding the exact solution for elementary shortest path problem, the pricing problem is still hard to solve, especially in large-scale instances. In this work, we further utilize the following

Algorithm 3

Edge-based branching procedure.

Input: path usages from a CG solution**Output:** two child nodes of the current node

- 1: Calculate edge usages based on path usages from a CG solution.
- 2: Select multiple candidate edges (10 edges in this study) with their usages closed to 0.5. Evaluate the impact of performing branching on these candidate edges as in [Dabia et al. \(2013\)](#). The quick pricing heuristic used in this study is the dynamic programming algorithm using heuristic domination and truncated labels introduced in the aforementioned acceleration techniques. Select an edge, say (i, j, ρ) , from all candidate edges such that branching on it results in the tightest estimated lower bound.
- 3: (Child node one) remove edge (i, j, ρ) from the graph used in the pricing problem.
- 4: (Child node two) remove edge set $\{(i', j', \rho') | (i', j', \rho') \in E_{NB}, i' = i\}$, edge set $\{(i', j', \rho') | (i', j', \rho') \in E_{NB}, j' = j\}$, and edge set $\{(i', j', \rho') | (i', j', \rho') \in E_{NB}, i' = j, j' = i\}$ from the graph used in the pricing problem, but keep edge (i, j, ρ) .
- 5: **return** child node one, child node two



Note: obj means the objective value of a BnB node; ns represents the solution state of a BnB node, it can be integer or fractional

Fig. 8. The overall framework of the BnB module.

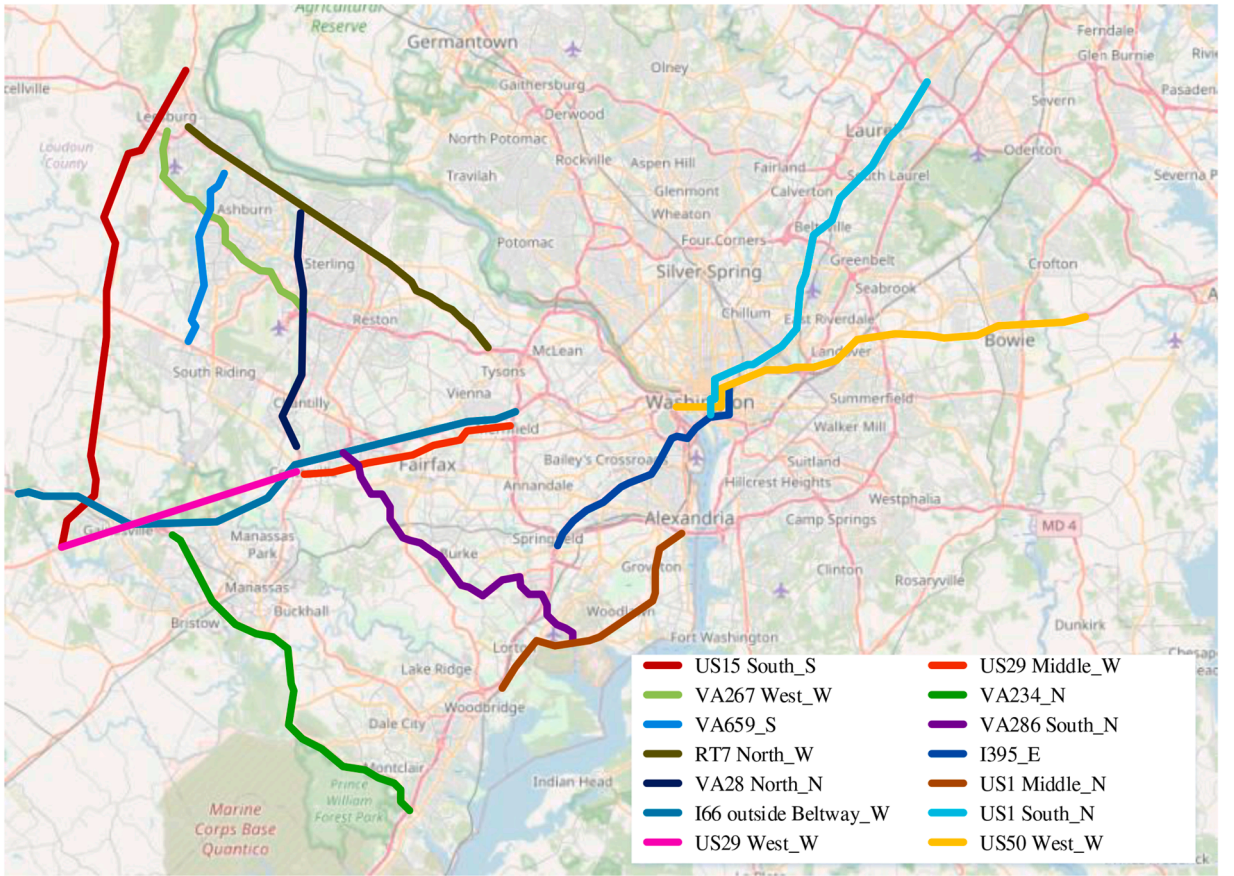


Fig. 9. Corridors used for time-dependent travel time modeling evaluation.

three techniques to speed up Algorithm 2.

Decremental state-space relaxation. It is observed that the larger ng-set of nodes are, the harder it is to solve the pricing problem. Therefore, instead of directly using the complete ng-set $\mathcal{V}_i$, Algorithm 2 can start with a subset of ng-set $\mathcal{V}_i$, i.e., $\mathcal{V}'_i$, for each node i and iteratively add nodes from $\mathcal{V}_i$ to $\mathcal{V}'_i$ until a valid ng-route is found. A node to be added to $\mathcal{V}'_i$ should satisfy the following requirement: the absence of that node in $\mathcal{V}'_i$ leads to a cycle in the final route that prevents the route from being an ng-route.

Heuristic domination. The first condition in the dominance rule presented above is by far stricter than the condition used in VRPs with the objective of minimizing total travel distance, which is $LR_c \leq l.R_c$. Using the condition $LR_c \leq l.R_c$ is of course helpful to dominate more labels, then accelerates the pricing process. However, some valid labels will also be discarded, which may affect the final solution. In our experiments, we found that only a small portion of valid labels are discarded due to the use of the loose domination condition. In other words, a relatively good solution can still be obtained with condition $LR_c \leq l.R_c$ in a much shorter time. Therefore, before using the exact dominance condition, i.e., $LR_c \leq l.R_c - (l.R_t - LR_t)$, the loose condition will be used until no valid route can be found.

Truncated labels. As in Yao et al. (2021), a truncated version of dynamic programming is used before calling the exact dynamic programming presented in Algorithm 2. In the truncated version, each node only maintains a limited number of promising labels, e.g., 100 labels with the least reduced cost. Note that, as the number of labels in the truncated version is limited, running one iteration of the truncated version is very fast, therefore we do not allow node revisiting and do not use heuristic domination such that routes with higher quality can be obtained.

Branch and bound

An optimal solution obtained from the CG module may contain fractional path usages, making it infeasible for the MP (model M3-CTR). Therefore, BnB module is further utilized to produce an integer optimal solution based on results from the CG module.

(a) Branching rule

If an optimal solution obtained from the CG module is fractional, the BnB module will perform branching on an edge with a fractional usage. The branching process is presented in Algorithm 3.

Child node one forces sprinkler trucks not to use edge (i, j, ρ) , while child node two forces sprinkler trucks to use edge (i, j, ρ) if

Table 6
Characteristics of the 12 corridors.

Corridor name	Type	# of links	Corridor length (mile)	LLS (mph)	HLS (mph)	ALS (mph)	ALCD (hour)
US15 South_S	Arterial	39	28.6	8.5	60.8	39.41	2.73
VA267 West_W	Freeway	20	13.0	16.6	74.7	63.80	0.49
VA659_S	Arterial	17	10.3	8.9	45.1	31.18	2.39
RT7 North_W	Arterial	36	19.7	14.0	62.1	44.76	3.68
VA28 North_N	Expressway	28	13.8	19.2	68.1	60.59	1.10
I66 outside Beltway_W	Freeway	50	27.9	15.0	74.7	58.47	2.93
US29 West_W	Arterial	27	13.1	14.0	55.9	42.19	2.82
US29 Middle_W	Arterial	17	11.8	11.5	51.2	26.45	6.76
VA234_N	Arterial	25	21.8	20.1	57.2	43.40	2.60
VA286 South_N	Arterial	27	19.1	14.4	58.1	47.15	1.40
I395_E	Freeway	28	9.7	6.1	65.8	43.26	5.63
US1 Middle_N	Arterial	14	13.7	8.8	52.7	36.88	4.74

Notes: HLS – highest link speed, LLS – lowest link speed, ALS – average link speed, ALCD – average link congestion duration for an entire day

Table 7
Comparison between observed and modeled speed across all links and 15-min interval resolution on 12 selected corridors.

Corridor name	Mean link MAE (mph)	Mean link MAPE (%)	Max link MAE (mph)
US15 South_S	2.77	8.83	4.77
VA267 West_W	2.86	4.52	3.22
VA659_S	2.38	8.48	3.62
RT7 North_W	4.09	11.04	5.91
VA28 North_N	2.47	4.55	5.03
I66 outside Beltway_W	4.35	9.10	7.52
US29 West_W	2.91	7.75	6.34
US29 Middle_W	4.02	16.92	7.64
VA234_N	3.20	8.14	7.53
VA286 South_N	3.05	7.47	5.94
I395_E	7.55	27.27	12.50
US1 Middle_N	3.41	11.08	6.99

Notes: MAE – mean absolute error, MAPE – mean absolute percentage error

service node i is in their routes.

(b) Searching strategy

A searching strategy determines the sequence of solving active BnB nodes, and different strategies may affect the performance of the BnB module for a specific problem. Breadth-first search (BFS) and depth-first search (DFS) are the two widely used searching strategies. Generally, BFS is helpful to improve the global lower bound (GLB) of a problem, while DFS is useful to quickly find a feasible solution thus contributes to improving the global upper bound (GUB) of the problem. Benefitting from mature parallel computing technologies, both BFS and DFS are implemented and run in parallel in this work. The overall framework of the BnB module designed in this paper is presented in Fig. 8.

From Fig. 8, in our implementation, in addition to performing BnB-DFS loop and BnB-BFS in parallel, as nodes in the same layer in BnB-BFS are independent from each other, solving nodes in Step 2b.1 is also conducted in a parallel manner.

6. Computational experiments

In this section, extensive numerical experiments are conducted to evaluate the methods developed in this paper. Specifically, 12 corridors in the Washington DC metropolitan area are first selected to demonstrate the suitability of the proposed time-dependent travel time modeling method. Based on calibrated travel time functions, three sets of SRP instances with different sizes are designed to examine the performance of the three models, followed by an analysis on the system impact of vehicle routings. Finally, sensitivity analysis of solution methods is performed.

6.1. Modeling of time-dependent travel time

Fig. 9 presents the 12 corridors selected for evaluating the proposed time-dependent travel time modeling method, including 1 expressway corridor, 3 freeway corridors, and 8 arterial corridors that experience different levels of traffic congestion. Table 6 summarizes the characteristics of the selected corridors. The time horizon of interest is set as 6:00 – 20:00. Note that transportation networks typically experience two or three congestion periods across a day (morning peak hours, afternoon peak hours, and possibly midday hours), while the method proposed in Section 2 is suitable for a single congestion period analysis. Hence, we split the analysis

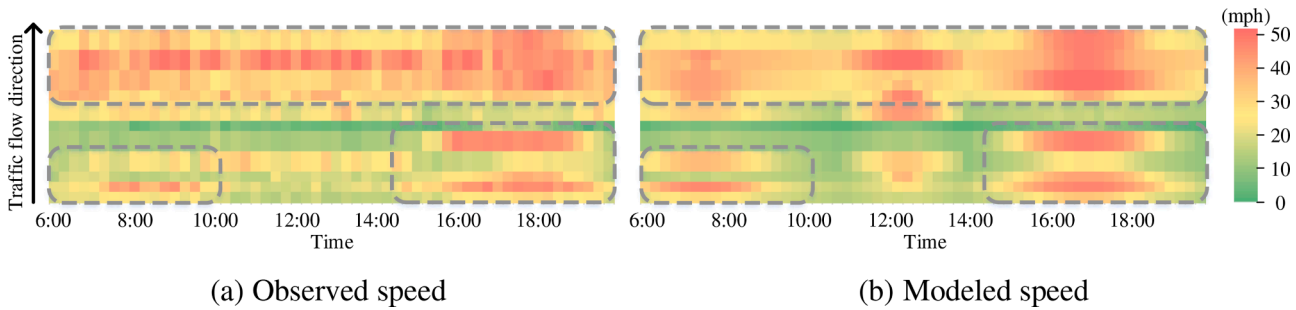


Fig. 10. Heatmap comparison between observed and modeled speed on corridor US29 Middle_W.

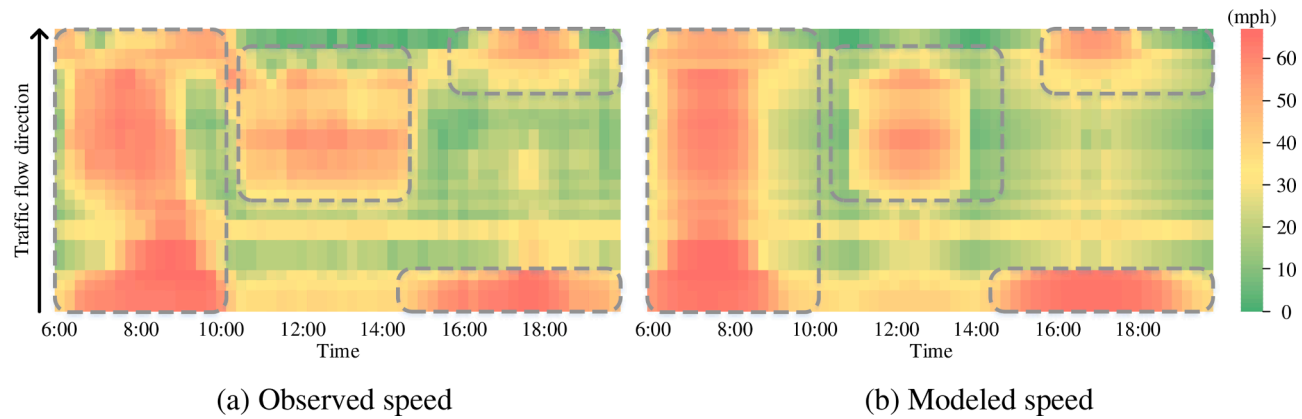
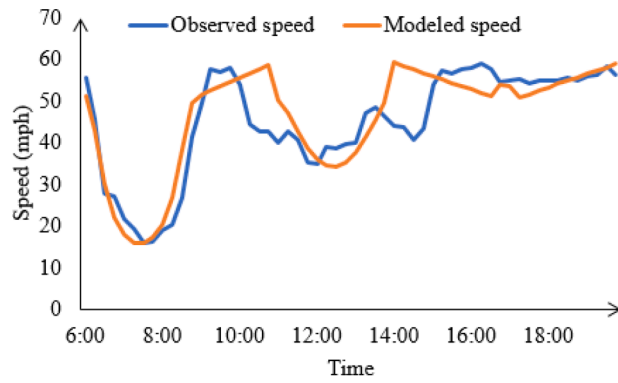
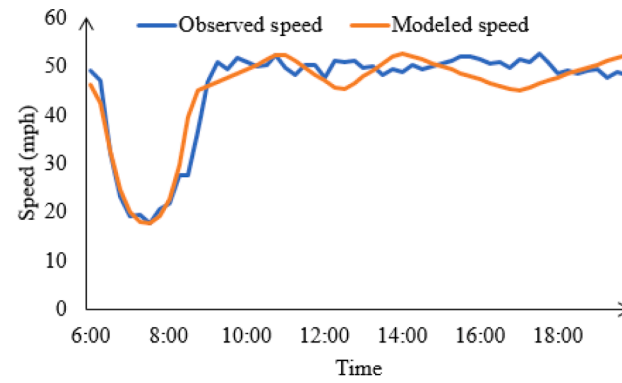


Fig. 11. Heatmap comparison between observed and modeled speed on corridor I395_E.



(a) Link 32609 on corridor I395_E



(b) Link 31948 on corridor US1 Middle_N

Fig. 12. Modeled speed and observed speed comparison on two links.

Table 8
Characteristics of three benchmark instance sets.

Small-size benchmark instance set				Medium-size benchmark instance set				Large-size benchmark instance set			
Instance	Links	MinLL	MaxLL	Instance	Links	MinLL	MaxLL	Instance	Links	MinLL	MaxLL
S1	16	0.19	0.53	M1	28	0.14	0.68	L1	72	0.04	0.68
S2	14	0.22	0.28	M2	36	0.11	0.65	L2	66	0.11	0.65
S3	16	0.09	0.29	M3	28	0.09	0.31	L3	60	0.09	0.54
S4	16	0.07	0.24	M4	30	0.14	0.42	L4	54	0.08	0.42
S5	16	0.13	0.50	M5	30	0.15	0.5	L5	60	0.13	0.5
S6	14	0.10	0.46	M6	38	0.13	0.44	L6	74	0.13	0.54
S7	18	0.18	0.50	M7	32	0.18	0.5	L7	62	0.18	0.63
S8	14	0.07	0.18	M8	34	0.07	0.49	L8	62	0.07	0.49
S9	16	0.13	0.37	M9	30	0.13	0.56	L9	50	0.13	0.56
S10	16	0.08	0.31	M10	28	0.15	0.33	L10	62	0.09	0.62
S11	15	0.09	0.68	M11	32	0.07	0.23	L11	60	0.07	0.7
S12	14	0.10	0.44	M12	30	0.09	0.41	L12	60	0.09	0.38
S13	18	0.09	0.23	M13	28	0.07	0.7	L13	58	0.07	1.43
S14	18	0.09	0.27	M14	32	0.09	0.37				
S15	16	0.15	0.74	M15	28	0.1	0.74				
Average	16	0.12	0.40	Average	31	0.11	0.49	Average	62	0.10	0.63

Notes: MinLL and MaxLL represent minimum link length and maximum link length respectively, with the unit of mile

time horizon into three periods, including AM (6:00–10:00), MD (10:00–14:00), and PM (14:00–20:00), then sequentially apply the method for each period. The dataset used for method evaluation was collected from the Regional Integrated Transportation Information System (RITIS). The raw dataset contains 5-minute space-mean traffic speed data from loop detectors and probe vehicles for a majority of links along each corridor. We first calibrate congestion duration P , average discharge rate μ , and inflow demand curvature parameter γ using the four-step method proposed in Section 2.1. Then smoothed time-dependent travel times are estimated based on Eq. (8) for each link along corridors.

Table 7 compares the observed speed and modelled speed calibrated from the fluid-queue model with polynomial arrival rates. The measure in terms of travel speed is selected as it is more intuitive than link travel time depending on a specific link length. First, reasonable modeling accuracy was achieved on all corridors with varied degrees of congestion. The heavily congested corridor I395_E has the highest mean link MAE (7.55 mph) and mean link MAPE (27.27%), as it has a relatively long congestion duration (5.63 hours for all three congestion periods during the entire day in Table 6). It should be remarked that, under a low average traffic speed associated with a heavy congestion, a typical deviation in absolute speed values could still result in a relatively large percentage difference, due to the smaller value in the denominator. In this regard, the mean link-based MAPE of corridor I395_E is still explainable. Second, for each corridor, max link MAE are not significantly worse than corresponding mean link MAE, indicating that the approximation error is reasonably bounded along each corridor. In short, better calibration results are observed on the corridors with a shorter congestion duration, whereas the corridors with severer congestion have larger modeling errors (e.g., corridor US29 Middle_W and I395_E).

To further examine the travel time modeling accuracy, Fig. 10 and Fig. 11 present the heatmap comparison between the observed and modeled travel speed on two corridors with a relatively poor modeling accuracy in Table 7, i.e., corridor US29 Middle_W and I395_E. On both corridors, the overall modeled speed pattern is closed to the corresponding observed one. Specifically, areas affected by congestion propagation (grey dash rectangles) have been realistically captured.

Focusing on two links with different traffic congestion patterns from corridor I395_E and corridor US1 Middle_N, Fig. 12 provides the time-dependent observed and modeled speed across the analysis time horizon. The congestion occurs in the periods of AM and MD on link 32609, whereas only am experiences significant delay on link 31948. One can observe that, for the link with a single congestion period (link 31948), the modeled speed closely matches the observed speed. However, for the link with multiple congestion periods (link 32609), there are still obvious deviations between two lines, which highlights the challenges in modeling complex and heavy traffic congestion.

6.2. Performance evaluation of the solution methods

As summarized in Table 8, three sets of instances with different sizes are designed to evaluate the three optimization models developed in this paper. Instance networks consist of expressway, freeway and arterial link. Travel time functions calibrated on the 12 representative corridors are applied to links in benchmark instances according to their road types. Our open-source repository can be found on Github at https://github.com/jiawlu/Time_Dependent_VRP_BnP.

The first instance set includes 15 small-size instances. The average number of road links to be cleaned is 16. The length of links in the first set ranges from 0.07 to 0.74 mile. The second set contains 15 medium-size instances, where the number of road links ranges from 28 to 38. The last set includes 13 large-size instances, with 62 links to be cleaned on average for each instance network. For all of the instances, the planning horizon is 3 hours. The vehicle capacity is 1200 gallons of water, and it is assumed that cleaning 1 mile of road consumes 400 gallons of water. The speed limit of sprinkler trucks in the cleaning mode and deadheading mode is set as 5 mph and 14 mph, respectively. We assume that there is no limit on sprinkler fleet size, but each sprinkler has an acquisition cost of

Table 9

Performance comparisons of three proposed models on small-size instances.

Instance	M1-TEN (LR)					M2-STD (MILP Solver)					M3-CTR (BnP)				
	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)
S1	108.01	112.6	4.08	136	0.14	23.34	112.38	79.23	900	22.33	112.38	112.38	0.0	1	0.0
S2	70.73	76.0	6.94	135	0.17	15.15	78.07	80.59	900	21.74	75.65	75.65	0.0	3	0.0
S3	65.96	73.4	10.13	540	0.54	10.0	73.03	86.31	900	22.46	73.03	73.03	0.0	20	0.0
S4	48.16	51.6	6.66	591	0.55	11.28	51.87	78.25	900	26.97	51.85	51.85	0.0	212	1.15
S5	77.64	82.6	6.01	315	0.28	12.59	83.24	84.87	900	22.68	82.36	82.36	0.0	5	0.0
S6	71.91	77.0	6.61	274	0.37	21.09	77.05	72.63	900	20.71	77.05	77.05	0.0	4	0.0
S7	120.59	132.0	8.64	284	0.29	15.32	131.87	88.38	900	21.72	131.87	131.87	0.0	6	0.0
S8	39.31	45.6	13.8	899	1.18	11.76	45.84	74.35	900	17.79	45.84	45.84	0.0	264	3.65
S9	74.91	82.6	9.31	293	0.24	13.24	80.28	83.5	900	13.67	80.02	80.02	0.0	4	0.0
S10	63.25	71.4	11.42	333	0.33	12.44	71.12	82.5	900	20.15	71.12	71.12	0.0	16	0.0
S11	121.22	130.8	7.32	900	0.94	62.99	130.69	51.8	900	19.88	130.69	130.69	0.0	1	0.0
S12	59.30	59.4	0.16	143	0.17	15.4	72.27	78.69	900	19.02	59.48	59.48	0.0	1	0.0
S13	68.94	78.0	11.62	227	0.26	11.32	78.9	85.65	900	27.19	78.53	78.53	0.0	62	6.83
S14	53.81	56.0	3.91	434	0.64	10.0	56.35	82.25	900	4.1	54.49	56.29	3.2	900	0.47
S15	109.57	111.4	1.65	273	0.37	26.66	113.35	76.48	900	26.75	111.4	111.4	0.0	1	0.0
Average			7.22	385	0.43			79.03	900	20.48			0.21	100	0.81

Table 10

Performance comparisons of three proposed models on medium-size instances.

Instance	M1-TEN (LR)					M2-STD (MILP Solver)					M3-CTR (BnP)				
	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)
M1	199.29	226.2	11.9	1265	0.98	151.12	212.76	28.97	1800	77.36	211.4	211.4	0.0	758	106.24
M2	215.73	245.8	12.23	1800	1.58	38.14	-	-	1800	17.23	221.28	221.28	0.0	27	0.0
M3	125.77	147.4	14.68	1182	1.08	10.0	146.64	93.18	1800	62.52	134.04	146.64	8.59	1800	156.71
M4	185.04	193.2	4.22	1800	1.11	53.32	193.07	72.39	1800	24.98	190.07	190.07	0.0	14	0.0
M5	217.37	232.4	6.47	1800	1.25	232.48	232.48	0.0	1186	9.19	232.48	232.48	0.0	4	0.0
M6	198.56	227.2	12.61	1800	1.35	10.0	253.17	96.05	1800	11.4	202.72	215.2	5.8	1800	169.99
M7	215.99	237.6	9.1	1800	1.09	15.2	-	-	1800	14.71	225.52	225.52	0.0	20	0.0
M8	132.51	160.4	17.39	1800	1.6	10.03	153.4	93.46	1800	28.81	136.06	150.12	9.37	1800	1.45
M9	185.2	201.2	7.95	1800	1.11	22.0	194.04	88.66	1800	35.02	189.89	192.76	1.49	1800	179.87
M10	148.68	156.0	4.69	1223	0.79	25.78	155.64	83.44	1800	24.82	153.03	153.03	0.0	22	0.0
M11	98.67	116.4	15.23	1800	1.7	10.0	112.2	91.09	1800	93.41	106.22	112.92	5.93	1800	4.62
M12	199.69	218.6	8.65	1245	0.92	13.29	222.67	94.03	1800	23.09	205.63	215.16	4.43	1800	132.7
M13	167.8	189.0	11.22	1800	1.27	10.02	186.37	94.62	1800	118.91	173.61	184.68	5.99	1800	175.27
M14	192.94	225.6	14.48	1800	1.1	10.0	197.49	94.94	1800	74.56	197.49	197.49	0.0	14	0.0
M15	198.52	225.0	11.77	1800	0.8	31.31	222.55	85.93	1800	30.59	204.3	222.55	8.2	1800	132.2
Average			10.84	1648	1.18			81.12	1759	43.11			3.32	1017	70.6

Table 11

Performance comparisons of three proposed models on large-size instances.

Instance	M1-TEN (LR)					M2-STD (MILP Solver)					M3-CTR (BnP)				
	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)
L1	203.71	632.0	67.77	3600	9.61	10.0	-	-	3600	25.26	531.0	543.87	2.37	3600	8.39
L2	159.6	540.4	70.47	3600	8.59	10.0	-	-	3600	21.36	450.57	467.53	3.63	3600	22.02
L3	112.4	629.6	82.15	3600	6.15	10.0	-	-	3600	13.5	553.44	558.19	0.85	3600	3.39
L4	10.0	346.8	97.12	3600	9.27	10.0	431.64	97.68	3600	19.71	330.26	341.46	3.28	3600	12.14
L5	10.0	789.8	98.73	3600	10.08	10.0	-	-	3600	15.8	371.99	403.93	7.91	3600	4.19
L6	10.0	647.0	98.45	3600	9.86	10.0	-	-	3600	17.99	465.71	504.8	7.74	3600	10.46
L7	69.64	1111.2	93.73	3600	8.29	186.99	-	-	3600	14.08	503.75	519.44	3.02	3600	19.36
L8	10.0	625.8	98.40	3600	9.50	10.0	-	-	3600	22.12	303.62	320.1	5.15	3600	7.55
L9	396.03	772.4	48.73	3600	5.32	105.18	-	-	3600	13.59	602.83	602.83	0.0	12	0.71
L10	27.06	637.4	95.76	3600	8.36	10.0	-	-	3600	16.8	382.14	399.41	4.32	3600	4.24
L11	10.0	694.0	98.56	3600	8.45	10.0	-	-	3600	17.55	320.05	338.0	5.31	3600	4.57
L12	162.28	1022.2	84.12	3600	7.55	10.0	-	-	3600	15.8	514.32	545.44	5.7	3600	23.17
L13	313.87	935.8	66.46	3600	5.32	10.0	-	-	3600	15.32	639.36	678.18	5.72	3600	15.72
Average			84.65	3600	8.18			99.82	3600	-			4.23	3324	10.45

Table 12

Optimality gaps of models M1-TEN and M3-CTR on large-size instances.

Instance	M1-TEN (LR)				M3-CTR (BnP)			
	LB	UB	Gap (%)	BestGap (%)	LB	UB	Gap (%)	BestGap (%)
L1	203.71	632.0	67.77	15.98	531.0	543.87	2.37	2.37
L2	159.6	540.4	70.47	16.62	450.57	467.53	3.63	3.63
L3	112.4	629.6	82.15	12.10	553.44	558.19	0.85	0.85
L4	10.0	346.8	97.12	4.77	330.26	341.46	3.28	3.28
L5	10.0	789.8	98.73	52.90	371.99	403.93	7.91	7.91
L6	10.0	647.0	98.45	28.02	465.71	504.8	7.74	7.74
L7	69.64	1111.2	93.73	54.67	503.75	519.44	3.02	3.02
L8	10.0	625.8	98.40	51.48	303.62	320.1	5.15	5.15
L9	396.03	772.4	48.73	21.95	602.83	602.83	0.0	0.0
L10	27.06	637.4	95.76	40.05	382.14	399.41	4.32	4.32
L11	10.0	694.0	98.56	53.88	320.05	338.0	5.31	5.31
L12	162.28	1022.2	84.12	49.68	514.32	545.44	5.7	5.7
L13	313.87	935.8	66.46	31.68	639.36	678.18	5.72	5.72
Average			84.65	33.37			4.23	4.23

equivalent 10 units of travel time. Note that, in this subsection, as we mainly focus on efficiency evaluation on the three proposed models, the weight of system impact cost ω is set as 0. The tradeoff between societal impact and private operating cost will be performed in [Section 6.3](#).

In the results presented below, model M1-TEN is solved by the LR method presented in [Section 5.1](#), M2-STD is directly solved by a commercial MILP solver (CPLEX), M3-CTR is solved by the BnP algorithm presented in [Section 5.2](#). The solution results of solving model M1-TEN directly by MILP solver (CPLEX) are also presented in [Appendix D](#) for reference. Both LR method and BnP algorithm proposed in this paper are coded in C++. The RMLP in BnP is solved with CPLEX by calling its built-in C++ API. We use CPLEX solver version 12.10 throughout the experiments. All numerical experiments conducted in this study are evaluated on a 64-bit Linux server with Intel Xeon Gold 6230R processor @ 2.10 GHz and 180 GB RAM.

The results of the proposed models on the three sets of real-life instances (small-size, medium-size, large-size) are shown in [Table 9](#), [Table 10](#), and [Table 11](#), respectively. For each instance, [Table 9 - Table 11](#) provide lower bound (LB), upper bound (UB), Gap, solution time (ST), and memory usage (MU) of results obtained from the proposed models on the three sets of instances, respectively. The Gaps in tables are relative gaps, which are calculated by $\text{Gap} = (\text{UB} - \text{LB})/\text{UB} \times 100$. The solution time limit is set as 15 minutes for small-size instances, 30 minutes for medium-size instances, 60 minutes for large-size instances. The time discretization resolution of model M1-TEN is set as 0.2 min. For model M3-CTR, the size of ng-set is set as 2, and the impact of different ng-set size is presented in [Appendix E](#).

Overall, the two models (M1-TEN and M3-CTR) solved by customized solution algorithms outperform the model (M2-STD) solved by a MILP solver in the three sets of instances; model M3-CTR has the best performance in terms of solution quality and solution time among the three proposed models.

For small-size instances, the average relative gaps of models M1-TEN, M2-STD, and M3-CTR are 7.22%, 79.03%, and 0.21%, respectively. Specifically, among the 15 small-size instances, model M3-CTR is able to produce optimal solutions on 14 instances within the time limit, while solution optimality cannot be proven on all 15 instances for models M1-TEN and M2-STD. On the other hand, although the relative gaps of model M1-TEN and M2-STD are higher than that of model M3-CTR, the upper bounds provided by the first two models are always closed to optimal solutions, indicating that these two models are still able to produce acceptable routing solutions for practical use in small-size instances. In terms of the computational speed, the average solution times of the three models on small-size instances are 385s, 900s, and 100s, respectively.

For medium-size instances, due to the increase of instance complexity, larger relative gaps and longer solution times are observed on three models. In terms of solution quality, similar to the results on small-size instances, model M3-CTR still has the best performance, with an average relative gap of 3.32%. For model M1-TEN, the average relative gap is 10.84%, while it is over 80% for model M2-STD. The large gap of model M2-STD is mainly due to its weak lower bounds. The model M2-STD is directly solved by CPLEX with built-in linear relaxation and branch-and-bound capability. Compared to model M3-CTR, we find that linearly relaxing a path-based model (M3-CTR) provides tighter lower bounds than simply relaxing an arc-based model (M2-STD).

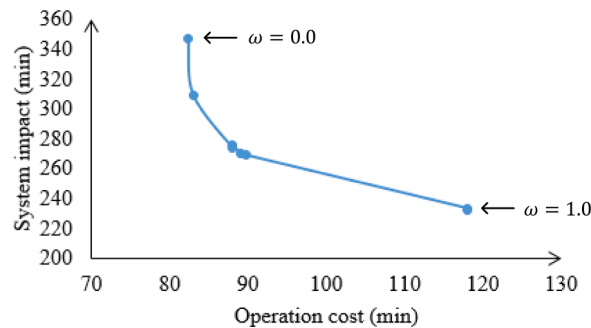
For large-size instances, model M2-STD is only able to produce feasible solutions on one instance (L4), and weak lower bounds are provided on all instances, resulting in an average relative gap of 99.82%. This highlights the need of designing customized solution algorithms for solving large-scale RAPPs in real life. Compared with model M3-CTR with an average gap of 4.23%, model M1-TEN provides relatively poor lower and upper bounds, resulting in an average gap of 84.65%. Its weak bounds are mainly due to the symmetry issue, as discussed by [Niu et al., \(2018\)](#) and [Yao et al., \(2019\)](#). As all sprinkler trucks are assumed to be identical, trucks tend to select the same least-cost path when serving customers, resulting in weaker lower bounds and additional efforts for generating feasible solutions for upper bounds.

In [Table 12](#), both Gap and BestGap are further provided to comprehensively measure the quality of solutions obtained from model M1-TEN and M3-CTR. BestGap represents the relative difference between upper bound and the best lower bound of the two models. The Gaps and BestGaps of model M3-CTR are exactly the same on all instances, while, for model M1-TEN, BestGaps are significantly smaller than the corresponding Gaps. The average BestGap of model M1 is 33.37%. This means that although the solution gaps of

Table 13Optimal routing costs of small instances under different weights of system impact ω .

Instance	Weight of system impact $\omega = 0.0$			–	Weight of system impact $\omega = 1.0$			OCPI (%)	SIPD (%)
	Total cost	Operation cost	System impact		Total cost	Operation cost	System impact		
S1	112.38	112.38	297.55		391.51	117.13	274.38	4.23	7.79
S2	75.65	75.65	320.44		335.83	88.63	247.20	17.16	22.85
S3	73.03	73.03	350.95		348.17	88.26	259.91	20.85	25.94
S4	51.85	51.85	328.06		328.10	87.81	240.29	69.35	26.76
S5	82.36	82.36	347.14		350.75	118.10	232.65	43.43	32.98
S6	77.05	77.05	289.92		318.01	89.25	228.76	15.83	21.09
S7	131.87	131.87	396.73		449.78	148.25	301.53	12.42	24.00
S8	45.84	45.84	318.53		292.49	58.60	233.89	27.84	26.57
S9	80.02	80.02	339.51		362.46	89.43	273.03	11.76	19.58
S10	71.12	71.12	343.32		339.80	88.73	251.07	24.76	26.87
S11	130.69	130.69	259.40		360.36	146.78	213.58	12.31	17.66
S12	59.48	59.48	278.47		297.54	89.42	208.12	50.34	25.26
S13	78.53	78.53	385.28		389.30	89.67	299.63	14.19	22.23
S14	-	-	-		-	-	-	-	-
S15	111.40	111.40	343.32		396.13	147.65	248.48	32.54	27.63

Notes: OCPI – operation cost percentage increase, SIPD – system impact percentage decrease

**Fig. 13.** Pareto curve of operation cost and system impact on instance S5 with ω ranging from 0.0 to 1.0.

model M1-TEN are relatively large, the upper bounds can actually serve as good feasible solutions in practice.

Another important finding from this comparison is the memory usage of each model, especially for real-time applications. Overall, model M1-TEN consumes much less memory than models M2-STD and M3-CTR. Even for large-size instances, the average memory usage of model M1-TEN is 8.18 GB, which highlight the applicability of model M1-TEN in on-line computing. Model M3-CTR consumes more memory than model M1-TEN. This is due to the use of parallel computing in the branch and bound module, which involves a large number of branch and bound nodes being processed simultaneously. For model M2-STD, the average memory usage on medium-size instances is 43.11 GB. Note that the memory statistics for model M2-STD was not reported for large-size instances, as it is still in the pre-solving stage when reaching the time limit. Another interesting finding is that the average memory usage of model M3-CTR on large-scale instances is less than that on medium-scale instances. This is due to the high complexity of large-scale instances, and only a few number of branch-and-bound nodes are explored within the pre-set time limit.

6.3. System impact and private cost of vehicle routing

Table 13 presents the optimal solution costs of small-size instances for $\omega = 0.0$ and 1.0. Note that, to exactly measure the changes of optimal operation cost and system impact under different weights of system impact, we only select instances that can be solved to optimality, thus instance S14 is not included in Table 13. Overall, after putting more priorities on the marginal cost of vehicle routing, the system impact obviously decreases while the private operation cost increases. For each instance, OCPI (SIPD) denotes the percentage increase (decrease) of the instance's optimal operation cost (system impact) when varying ω from 0.0 to 1.0. Due to the diversity of selected instances, OCPI and SIPD may vary from one instance to another. Specifically, for instance S1, its operation cost increases by 4.23% when changing $\omega = 0.0$ from 0.0 to 1.0, while it increases by 69.35% in instance S4.

As an example, Fig. 13 presents the Pareto frontier of operation cost and system impact on instance S5 with ω ranging from 0.0 to 1.0. Thus, a pareto-optimal solution should be systematically selected by decision makers and planners to balance these two important criteria.

7. Conclusions

In this study, focusing on formulating and solving RARPs in city logistics with a congested urban environment, we propose a comprehensive modeling framework and exact solution algorithms. Specifically, based on the fluid queuing model with a polynomial functional assumption for arrival flow rates, time-dependent link travel time as well as system-wide (societal) congestion impact is analytically derived. Two new time-dependent travel time representation schemes are investigated. With the SRP as an example, three optimization models, including a time-expanded network based arc routing formulation (model M1-TEN), an arc-based node routing formulation (model M2-STD), and a path-based node routing formulation (model M3-CTR), are constructed from different perspectives of capturing time-dependent travel time and formulating problem-specific constraints. In addition, two exact solution methods, i.e., Lagrangian relaxation and branch-and-price algorithm, are developed for efficiently solving model M1-TEN and model M3-CTR.

With real-world traffic flow data collected from the Washington DC metropolitan area, 12 representative corridors are selected to demonstrate the effectiveness of the proposed time-dependent link travel time modeling method. Results show that the proposed method can reasonably capture travel time evolution profiles under different levels of traffic congestion. Based on calibrated travel time functions, three sets of SRP instances with different sizes are designed to examine the performance of the three optimization models and their corresponding solution algorithms. It is found that, overall, model M3-CTR produces high-quality solutions in a shorter time, while model M1-TEN is able to provide good feasible solutions with less memory consumption. Furthermore, routing solutions with different priorities on system-wide (societal) congestion impact are systematically analyzed.

This study can be extended along the following directions in the future. First, this paper considers fixed departure time of service vehicles at the origin depot. Future studies could relax this restriction to allow the selection of proper departure times/schedules so as to minimize the overall cost. One interesting yet challenging topic along this line is how to produce optimal continuous-time path solutions without explicitly performing time discretization. This theoretically important and practically useful question was discussed in a recent paper by Boland et al., (2017) for the service network design problem. Second, with simplified fluid queue models, the system-wide (societal) congestion impact of routing solutions is considered in this paper. The tradeoff among additional costs such as energy use, emissions reduction, and location congestion reduction (Lam and Hentenryck, 2016) could be further systematically investigated under more realistic traffic flow modeling frameworks to accommodate different practical needs in real-life applications. Third, time-dependent travel time is modeled with a deterministic form in this study, which may result in sub-optimal vehicle routing solutions in real-life applications, due to the stochasticity of travel times. A stochastic time-dependent travel time model can be further developed and incorporated into vehicle routing models in future studies to produce robust solutions. Specifically, we could consider a possible reformulation approach, based on a Lagrangian relaxation framework proposed by Yang and Zhou (2014), to take into account multiple stochastic scenarios due to stochastic capacity or demand factors.

CRedit authorship contribution statement

Jiawei Lu: Methodology, Writing – original draft, Writing – review & editing. **Qinghui Nie:** Conceptualization, Data curation, Writing – original draft. **Monirehalsadat Mahmoudi:** Conceptualization, Methodology. **Jishun Ou:** Conceptualization, Writing – original draft. **Chongnan Li:** Validation, Writing – review & editing. **Xuesong Simon Zhou:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

Appendix A. Graph transformation

The key idea of converting the original ARP to an NRP is treating each link to be cleaned as a customer (activity node) and connecting different customers with virtual edges. In Fig. A1, we use an illustrative network with 4 intersections and 8 directed road links as an example to introduce the process of converting the original ARP to an NRP. Among the four intersections in Fig. A1(a), intersection 1 and 3 also serve as the depot of sprinkler trucks and water refilling station, respectively. The workflow of transforming Fig. A1(a) to Fig. A1(b) is presented in Algorithm A1.

In Step 3, the service time of service node i ($\tau_i(t)$) equals to the cleaning time of its corresponding road link. That is, $\tau_i(t) = CT(t)$, where $CT(t)$ is the link cleaning time in Eq. (16).

In Step 5, up to φ edges will be constructed between each pair of activity nodes, with each edge corresponding to a path in the physical network. Paths can be obtained from candidate path sets in real life or searched by k-shortest path algorithms (e.g., Scano et al., 2015; Lu et al., 2018). Time-dependent path travel time and system impact can be calculated using the method proposed in Section 2.

To reduce the size of the resulting graph as much as possible, we only build edges that is possibly used in an optimal solution. For example, as we require that sprinkler trucks are full of water when departing from the origin depot, an optimal solution will not contain an edge from the origin depot to a water refilling station. Therefore, edges from the origin node to water refilling stations will not be built.

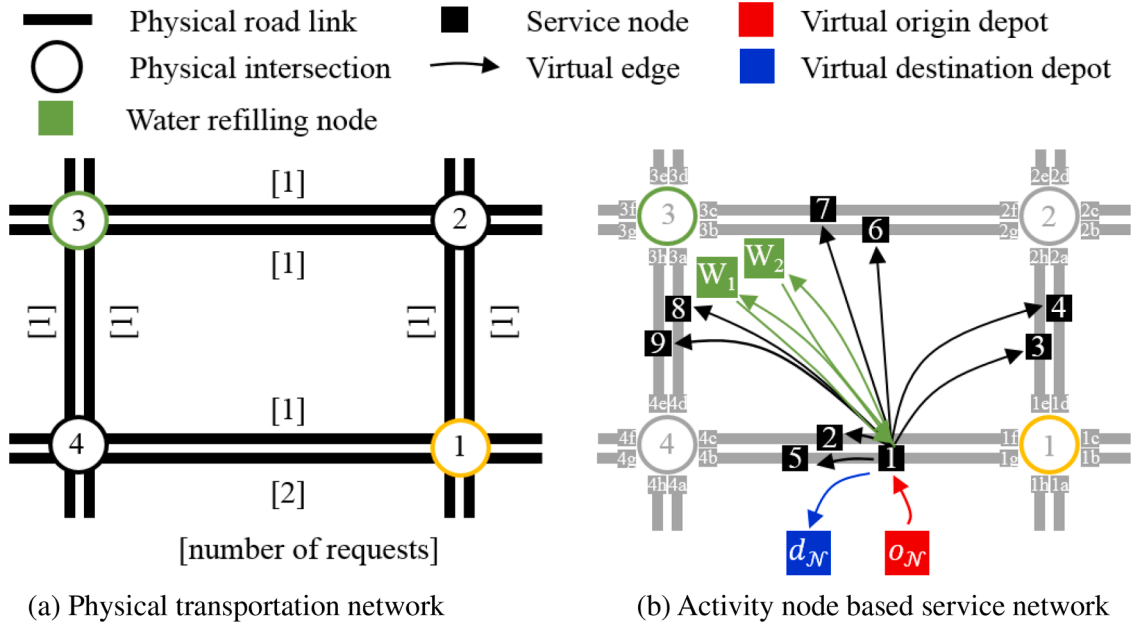


Fig. A1. A simple transportation network and its activity node based service network.

Algorithm A1

Graph transformation process.

Input: physical transportation network G

Output: activity node based network $G_{\mathcal{F}} = (N_{\mathcal{F}}, E_{\mathcal{F}})$, node service time function $\tau_i(t), i \in N_{\mathcal{F}}$, edge travel time function $\tau_{ij,p}(t), (i, j, p) \in E_{\mathcal{F}}$

- 1: **(Prepare Underlying Network)**
- 2: Step 1: Build an intersection-expanded underlying network from the physical network
- 3: Build an intersection-expanded underlying network $G_{\mathcal{F}}$ (the network in grey in Fig. B1(b)) from Fig. B1(a).
- 4: **(Construct Activity Node Set $N_{\mathcal{F}}$)**
- 5: Step 2: Construct origin depot and destination depot
- 6: Create a virtual origin depot $o_{\mathcal{F}}$ and a virtual destination depot $d_{\mathcal{F}}$ from intersection 1.
- 7: Step 3: Construct service nodes
- 8: Create service nodes for each road link with service requests in $G_{\mathcal{F}}$. The number of service nodes created for a specific road link should be equal to its number of service requests. We build two service nodes (node 1 and node 5) for road link (4,1) with two cleaning requests, and one service node for each other road link.
- 9: Step 4: Construct water refilling nodes
- 10: Create dummy water refilling nodes for each intersection that serves as water refilling station, e.g., node W_1 and W_2 for inter section 3 in our example. Dummy water refilling nodes are created to avoid multiple visits at one water refilling station. The number of dummy water refilling nodes needed for each physical water refilling station equals $\text{ceiling}(\text{total water needed to complete all cleaning tasks} / \text{water tank capacity of sprinkler trucks})$.
- 11: **(Construct Edge Set $E_{\mathcal{F}}$)**
- 12: Step 5: Construct virtual edges between activity nodes
- 13: $E_{\mathcal{F}} = \{(i, j, p) | i \in N_{\mathcal{F}}, j \in N_{\mathcal{F}}, p \in \{1, 2, \dots, \varphi\}\}$. For each service node, build virtual links with origin depot, destination depot, other service nodes, and water refilling stations. As an example, we show the virtual edges created for service node 1 in Fig. B1(b).
- 14: **return** $G_{\mathcal{F}} = (N_{\mathcal{F}}, E_{\mathcal{F}})$

Appendix B. Model M2 with nonlinear functions

Model M2 with nonlinear functions is presented as follows, where notations are illustrated in Table B1.

Objective function

$$\min Z_2 = \sum_{f \in F} f_{d_{\mathcal{F}}}^d + h \sum_{(j,p):(o_{\mathcal{F}},j,p) \in E_{\mathcal{F}}} x_{o_{\mathcal{F}},j,p} + \omega \left(\sum_{(i,j,p) \in E_{\mathcal{F}}} c g_{i,j,p}(t_i^d) x_{i,j,p} + \sum_{(i,j,p) \in E_{\mathcal{F}}} c g_j(t_j^s) x_{i,j,p} \right) \quad (\text{B1})$$

Subject to:

Cleaning request satisfaction constraint:

$$\sum_{(j,p):(j,i,p) \in E_{\mathcal{F}}} x_{j,i,p} = 1, \forall i \in N_{\mathcal{F}}^s \quad (\text{B2})$$

Sprinkler spatial route constraint:

Table B1
Notation list.

Symbols	Definition
Indices	
i, j	Index of nodes in graph $G_{\mathcal{N}}$
ρ	Index of edges between each pair of nodes in graph $G_{\mathcal{N}}$
Sets	
$N_{\mathcal{N}}$	Set of nodes in graph $G_{\mathcal{N}}$
$N_{\mathcal{N}}^s$	Set of service nodes in graph $G_{\mathcal{N}}$
$N_{\mathcal{N}}^w$	Set of water refilling stations in graph $G_{\mathcal{N}}$
$E_{\mathcal{N}}$	Set of edges in graph $G_{\mathcal{N}}$
Functions	
$\tau_i(t)$	Time-dependent service time function of node $i \in N_{\mathcal{N}}$
$\tau_{i,j,\rho}(t)$	Time-dependent travel time function of edge $(i,j,\rho) \in E_{\mathcal{N}}$
$cg_i(t)$	Time-dependent system impact function of serving node i at time t
$cg_{i,j,\rho}(t)$	Time-dependent system impact function of using edge (i,j,ρ) at time t
Parameters	
φ	Number of edges between each pair of nodes in graph $G_{\mathcal{N}}$
w_i	Water consumption of node $i \in N_{\mathcal{N}}$
$s_i(e_i)$	The earliest (latest) service starting time of node $i \in N_{\mathcal{N}}$
Variables	
$x_{i,j,\rho}$	Binary variable. $x_{i,j,\rho} = 1$ if a sprinkler uses edge (i,j,ρ) ; otherwise, $x_{i,j,\rho} = 0$
q_i	Water level of a sprinkler at node i
t_i^d	Departure time of a sprinkler at node i
t_i^s	Service starting time of node i

$$\sum_{(j,\rho):(j,i,\rho) \in E_{\mathcal{N}}} x_{j,i,\rho} = \sum_{(j,\rho):(i,j,\rho) \in E_{\mathcal{N}}} x_{j,i,\rho}, \forall i \in N_{\mathcal{N}}^s \cup N_{\mathcal{N}}^w \quad (\text{B3})$$

$$\sum_{(j,\rho):(j,i,\rho) \in E_{\mathcal{N}}} x_{j,i,\rho} \leq 1, \forall i \in N_{\mathcal{N}}^w \cup \{d_{\mathcal{N}}^f | f \in F\} \quad (\text{B4})$$

Sprinkler temporal route constraint:

$$t_{o_{\mathcal{N}}}^d = 0 \quad (\text{B5})$$

$$t_j^s \geq t_i^d + \tau_{i,j,\rho}(t_i^d) + M(x_{i,j,\rho} - 1), \forall (i,j,\rho) \in E_{\mathcal{N}} \quad (\text{B6})$$

$$t_i^d \geq t_i^s + \tau_i(t_i^s), \forall i \in N_{\mathcal{N}} \quad (\text{B7})$$

Time window constraint:

$$s_i \leq t_i^s \leq e_i, \forall i \in N_{\mathcal{N}} \quad (\text{B8})$$

Sprinkler truck water level updating constraint:

$$q_{o_{\mathcal{N}}} = C \quad (\text{B9})$$

$$q_j \leq q_i - x_{i,j,\rho} w_j + C(1 - x_{i,j,\rho}), \forall (i,j,\rho) \in \{E_{\mathcal{N}} | i \in N_{\mathcal{N}}^s\} \quad (\text{B10})$$

$$q_j \leq C - x_{i,j,\rho} w_j, \forall (i,j,\rho) \in \{E_{\mathcal{N}} | i \in N_{\mathcal{N}}^w \cup \{o_{\mathcal{N}}\}\} \quad (\text{B11})$$

Decision variables:

$$x_{i,j,\rho} \in \{0, 1\}, \forall (i,j,\rho) \in E_{\mathcal{N}}$$

$$0 \leq q_i \leq C, t_i^d \geq 0, t_i^s \geq 0, \forall i \in N_{\mathcal{N}} \quad (\text{B12})$$

Appendix C. Lagrangian relaxation based solution process for solving M1-TEN

Algorithm C1

Lagrangian relaxation based solution process.

Step 1: Initialization

Initialize iteration number $k = 0$;

Initialize Lagrangian multipliers λ_{ij}^k ;

Initialize the best lower bound $LB^* = -\infty$ and the best upper bound $UB^* = +\infty$.

Step 2: Search paths for vehicles

Call dynamic programming algorithm once to find the best paths for origin to destination on the time-expanded network with arc cost $c'_{i,j,t,t'}$ in Eq. (44).

Choose the best $|F|$ paths with the minimum reduced costs from the origin to destination returned by the dynamic programming algorithm.

Assign the $|F|$ paths to $|F|$ vehicles. Note that if the reduced cost of a path is larger than zero, the corresponding vehicle will not use the path and will keep staying at the depot.

Step 3: Update lower bound

$LB^k =$ sum of reduced costs of vehicle paths minus $\sum_{(i,j) \in L_r} \lambda_{ij} m_{ij}$ in Eq. (44).

$LB^* = \max\{LB^*, LB^k\}$

Step 4: Update upper bound

Check the service node to vehicle assignment in the lower bound solution. If a service node is served by more than one vehicle, designate one of the vehicles for it.

If a service node is not served by any vehicle, assign a virtual vehicle for it. The above operations result in a feasible solution

$UB^k =$ sum of travel time costs of vehicle paths plus vehicle acquisition costs in the above feasible solution

$UB^* = \min\{UB^*, UB^k\}$

Step 5: Evaluate the solution quality and termination condition

$GAP = (UB^* - LB^*)/UB^* \times 100\%$

If time limit is reached or the GAP does not change in the last 20 iterations, iteration stops. Otherwise, update Lagrangian multipliers using Eqs. (45) and (46), $k = k + 1$, go back to Step 2.

Appendix D. Solution results of model M1-TEN using MILP solver (CPLEX)

The solution results of model M1-TEN using CPLEX are presented in Table D1.

Table D1

Solution results of the three sets of instances (M1-TEN, CPLEX)

Instance	Small-size instance set					Medium-size instance set					Large-size instance set				
	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)	LB	UB	Gap (%)	ST (s)	MU (GB)
1	112.6	112.6	0.0	70	4.14	200.83	212.4	5.45	1800	17.5	476.47	-	-	3600	16.44
2	71.23	75.8	6.03	900	7.22	217.02	233.2	6.94	1800	23.66	398.46	-	-	3600	33.66
3	66.28	73.2	9.45	900	7.57	126.71	150.4	15.75	1800	2.09	545.72	-	-	3600	17.74
4	48.32	51.6	6.35	900	7.29	186.28	191.4	2.67	1800	17.98	326.4	-	-	3600	14.89
5	80.34	82.4	2.5	900	6.58	228.63	232.4	1.62	1800	16.36	372.19	-	-	3600	19.7
6	73.11	77	5.05	900	6.58	200.29	-	-	1800	20.23	427.87	-	-	3600	22.84
7	131.8	131.8	0.0	827	8.48	216.86	225	3.61	1800	18.27	439.56	-	-	3600	20.82
8	40.69	45.6	10.76	900	7.51	134.81	159	15.21	1800	20.44	275.43	-	-	3600	18.08
9	79.8	79.8	0.0	798	6.89	185.16	193.8	4.46	1800	17.9	602.2	-	-	3600	13.39
10	62.99	71.4	11.77	900	7.99	150.58	153.4	1.84	1800	18.31	325.81	-	-	3600	19.45
11	125.3	131	4.35	900	7.39	102.29	114.2	10.43	1800	16.97	303.84	-	-	3600	15.99
12	59.4	59.4	0.0	6	0	200.65	218.2	8.04	1800	16.94	512.13	-	-	3600	15.02
13	69.7	78.6	11.32	900	7.04	168.96	187.8	10.03	1800	16.89	619.23	-	-	3600	13.32
14	54.34	56	2.96	900	9.44	195.23	236.4	17.42	1800	9.42					
15	111.4	111.4	0.0	73	5.61	197.14	224.8	12.31	1800	17.06					
Average			4.70	718	6.65			14.39	1800	16.67			100.0	3600	18.56

Appendix E. Impact of the ng-set size on the solutions from CG module

In general, the larger ng-set is, the tighter lower bounds can be obtained, at the same time, also harder to solve. In Table E1, with different sizes of ng-set, CG results for the LMPs of all instances are presented. ngx means the size of ng-set is x. In our implementations, we choose x nearest nodes (Euclidean distance) as the ng-set for each node. OBJ and ST represent the objective value of LMPs and solution time, respectively. “-” means no solution is obtained within the time limit, which is 7200 seconds. An obvious finding from Table E1 is that, for each instance, OBJs with a larger ng-set are higher than or equal to those with a smaller ng-set. The reason is that larger ng-sets prevent more cycles; thus, tighter lower bounds can be obtained. In terms of solution time, a trend that solution time increases with the increase of ng-set size can be observed. This trend is more obvious in large-scale instances. Specifically, for instances L5 and L11, they are no longer solvable within the preset time limit when the size of ng-set is larger than 4 and 8, respectively. In this paper, considering the tradeoff between computation efficiency and lower bound quality of the pricing problem, we set the size of ng-sets as 2 for all instances.

Table E1

Results of CG with different size of ng-sets.

Instance	ng1		ng2		ng4		ng8		ng16		ng24	
	OBJ	ST (s)	OBJ	ST (s)	OBJ	ST (s)	OBJ	ST (s)	OBJ	ST (s)	OBJ	ST (s)
S1	108.029	0.06	108.029	0.06	108.029	0.08	108.029	0.08	108.029	0.09	108.029	0.06
S2	69.888	0.09	70.0238	0.09	70.9043	0.1	70.9043	0.1	70.9043	0.1	70.9043	0.07
S3	65.288	0.37	65.31	0.77	65.328	0.42	65.328	0.42	65.6978	0.72	66.0244	1.33
S4	48.3182	6.26	48.3409	15.94	48.5236	5.21	48.6318	8.39	48.8493	40.4	48.9767	87.92
S5	78.44	0.18	78.44	0.18	78.44	0.28	78.44	0.25	78.9467	0.45	78.9467	0.49
S6	71.296	0.19	71.296	0.2	71.296	0.15	71.6471	0.25	72.04	0.32	72.04	0.41
S7	121.195	0.07	121.239	0.09	121.239	0.08	121.239	0.08	121.239	0.07	121.239	0.1
S8	39.0128	113.87	39.0796	28.4	39.0831	32	39.2578	24.71	39.972	2059.19	39.972	1899.03
S9	74.8906	0.19	75.156	0.24	75.173	0.18	75.1946	0.21	75.29	0.27	75.29	0.23
S10	62.1868	0.34	62.8888	0.39	62.8888	0.3	63.2029	0.67	63.5043	0.86	63.5043	0.84
S11	123.032	0.05	123.032	0.05	123.032	0.03	123.032	0.03	123.032	0.05	123.032	0.03
S12	59.48	0.34	59.48	0.29	59.48	0.37	59.48	0.59	59.48	0.72	59.48	0.72
S13	69.5325	0.5	69.5589	0.68	70.0774	0.56	70.0774	0.57	70.0774	0.59	70.0774	0.55
S14	54.4759	1646.83	54.4881	523.45	54.5064	646.67	54.5242	1390.39	-	-	54.7112	7130.58
S15	108.9	0.09	108.9	0.1	110.216	0.14	110.216	0.11	110.216	0.11	110.216	0.11
M1	200.91	0.43	201.281	0.74	201.406	0.83	201.406	0.88	201.449	1.19	201.491	0.76
M2	217.711	1.08	217.738	1.13	217.775	1.62	217.775	1.88	217.775	1.71	217.775	1.9
M3	127.228	31.31	128.768	25.76	129.268	18.16	129.712	41.08	129.719	49.23	129.719	56.68
M4	184.893	0.62	184.968	0.62	184.979	0.62	184.979	0.58	184.979	0.59	184.979	0.57
M5	232.48	0.34	232.48	0.34	232.48	0.49	232.48	0.49	232.48	0.51	232.48	0.48
M6	199.284	2.77	199.558	6.16	200.201	7.56	200.21	5.16	200.346	11.78	200.449	12.52
M7	219.806	0.66	219.838	0.65	220.111	0.92	220.111	0.91	220.111	0.98	220.111	0.95
M8	135.22	618.23	135.464	892.36	135.684	629.39	135.739	1030.65	136.315	3605.41	136.315	3884.25
M9	185.176	0.95	185.525	0.94	185.972	1.67	186.36	1.45	186.535	2.03	186.535	1.99
M10	149.708	2.42	149.708	3.43	149.715	2.54	149.803	1.99	149.836	1.93	149.855	1.83
M11	103.546	94.54	103.878	305.52	104.189	210.63	104.189	281.04	104.189	306.43	104.205	232.02
M12	199.124	0.5	199.128	0.62	199.128	0.66	199.297	0.81	199.733	0.99	199.733	0.98
M13	168.924	2.18	169.093	6.02	169.105	6.27	169.111	7.05	169.111	7.42	169.137	7.1
M14	195.959	3.13	195.959	2.4	196.473	2.77	196.473	2.98	196.473	3.3	196.473	3.31
M15	198.034	1.01	198.163	1.16	199.476	1.08	199.48	1.06	199.523	1.63	200.763	1.58
L1	530.124	607.56	530.151	795.78	530.269	809.19	530.269	586.33	530.269	578.46	530.429	654.29
L2	448.089	117.5	448.668	170.79	448.926	373.43	448.932	487.83	448.932	501.66	448.932	446.07
L3	549.765	1147.19	550.748	1345.2	550.748	1438.24	551.303	1144.81	551.321	2162.1	551.321	1245.25
L4	326.572	701.45	326.572	620.78	326.572	562.34	326.572	637.25	326.572	938.29	326.572	534.64
L5	371.434	180.62	371.993	272.2	-	-	-	-	-	-	-	-
L6	464.255	199.29	464.79	494.17	465.996	620.48	466.142	671.9	466.22	521.86	466.22	634.64
L7	499.571	47.83	501.104	119.58	501.523	137.9	501.525	123.07	501.579	146.96	501.579	148.16
L8	302.927	173.68	303.094	535.89	303.245	654.31	303.256	1116.18	303.509	660.22	303.587	852.03
L9	602.83	15.14	602.83	15.18	602.83	14.81	602.83	14.87	602.83	16.07	602.83	15.25
L10	382.122	1943.08	382.141	3373.92	382.256	3312.17	382.258	2922.25	-	-	382.311	6728.26
L11	319.703	1480.31	320.049	1993.82	320.288	4319.85	-	-	-	-	-	-
L12	514.324	204.71	514.324	181.39	514.324	177.38	514.324	180.85	514.324	175.39	514.324	180.69
L13	633.217	156.12	633.317	172.75	637.141	938.58	637.396	1201.52	637.473	1752.91	638.252	1389.4

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